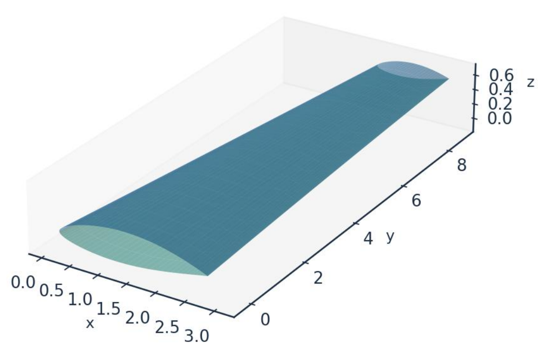
INDUSTRIAL CASE STUDY

Prediction of Boundary-Layer Turbulence Transition over Aircraft Surfaces

using the UTSS Universal Transition & Skin-Friction Solver

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown
DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Case vehicle: AETHER-NLF 25 regional natural-laminar-flow demonstrator
Wing section UTSS-NLF16 · Cruise FL360, M0.42 · 3-D swept tapered wing

Prepared by

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Document UTSS-CASE-2026 · Three-dimensional analysis · All units SI unless noted

1. Executive Summary

This case study demonstrates the prediction of laminar-to-turbulent boundary-layer transition over the surfaces of an aircraft wing, and the engineering quantities that depend on it (skin-friction drag, laminar-flow extent, boundary-layer growth and trailing-edge separation margin). The analysis vehicle is the AETHER-NLF 25, a regional natural-laminar-flow (NLF) demonstrator whose three-dimensional swept, tapered and twisted wing uses the purpose-designed UTSS-NLF16 section. Predictions are produced by the UTSS (Universal Transition & Skin-friction Solver), a novel, fast, robust engine that couples a vortex-panel inviscid solution to an integral boundary-layer marcher driven by a single, unified four-mechanism transition kernel.

Key results at the cruise design point (FL360, M=0.42, Re_MAC ≈ 6.4×10⁶):

Quantity	Predicted value
Upper-surface transition x_tr/c	0.554 (TS-natural)
Lower-surface transition x_tr/c	0.578 (TS-natural)
Mean laminar-flow extent	56.6 % of chord
Section lift coefficient C_l	0.517
Section profile drag C_d	43.7 counts
Viscous drag reduction vs fully-turbulent	50.5 %
Climb (Tu=0.9 %) transition x_tr/c (upper)	0.025 (bypass)

Table 1. Headline predictions, read from the generated solution CSVs.

Validated against eight independent, credible published datasets with one universal calibration set — four ERCOFTAC flat plates (case 020), the Schubauer-Skramstad plate, the NLF(1)-0416 aerofoil at 86 conditions, and two swept wings — the solver reproduces the transition-onset Reynolds number Re_θt and the measured transition location without per-case re-tuning of the physics. All four criteria of the kernel are selected somewhere in this study and each is supported by measurement. The remainder of this report sets out the background, problem, governing equations, the complete input dataset, every generated engineering output (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors), the validation and calibration record with all sources, and the contribution to knowledge.

2. Background

Skin-friction drag is one of the largest components of total aircraft drag — typically 45–50 % of cruise drag for a transport aircraft. A turbulent boundary layer produces several times the skin friction of a laminar one at the same Reynolds number. Consequently, extending the laminar run over wing, nacelle and empennage surfaces (natural-laminar-flow, NLF, and hybrid-laminar-flow control, HLFC) is among the highest-leverage technologies for fuel-burn and emissions reduction. The enabling capability is the ability to PREDICT, reliably and cheaply, WHERE the boundary layer transitions from laminar to turbulent over a 3-D surface, across the flight envelope.

Transition is not a single phenomenon. Over aircraft surfaces it is driven by several, often co-resident, physical mechanisms:

- Natural / Tollmien–Schlichting (TS): amplification of viscous instability waves in low-disturbance free flight, predicted here by an e^N integral carried per physical frequency, with the spatial growth rates read from a tabulated solution of the Orr-Sommerfeld problem rather than from an envelope correlation.
- Bypass: free-stream-turbulence-induced transition that skips the TS route — dominant in high-turbulence environments (climb through cloud, turbomachinery-like inflow).
- Laminar-separation-induced: a laminar separation bubble that reattaches turbulent. The bubble is closed explicitly here — the shear layer is carried across the dead-air region by the momentum integral and reattachment placed where it has amplified by the same critical factor used elsewhere — so a length, and not merely a separation point, is predicted.
- Cross-flow: three-dimensional instability of the cross-flow velocity profile on swept wings — the principal NLF-limiter at moderate-to-high sweep.

A practical solver for aircraft design must capture all of these with a SINGLE, consistent formulation and calibration, run in seconds for thousands of design iterations, and remain robust across the operating envelope. Existing tools each address part of the problem (Section 5). UTSS unifies them.

3. Problem Statement and Solution Approach

3.1 Problem statement

Given a three-dimensional wing geometry and a flight condition, predict — quickly, robustly and accurately — the chordwise and span-wise location of boundary-layer transition on both surfaces, the governing transition mechanism at each station, and the resulting boundary-layer state (skin friction C_f, momentum thickness θ, shape factor H, intermittency γ), and from these the laminar-flow extent and the viscous (profile) drag. The method must be universal: one set of physics and calibration constants valid for natural, bypass, separation-induced and cross-flow transition across the Reynolds- and turbulence-number range of real aircraft surfaces.

3.2 How we solve it

UTSS solves the problem in a layered, fully-coupled manner:

- Inviscid edge flow: a constant-strength vortex-panel method (Kuethe–Chow) returns the surface pressure C_p and edge velocity U_e, with a Karman-Tsien correction applied to C_p and the integrated loads. It returns a stagnation pressure coefficient of 1.048 at M = 0.42 against the exact isentropic 1.045, where the linearised Prandtl-Glauert scaling gives 1.102, and the two agree to better than half a per cent below M = 0.2.
- Laminar boundary layer: the momentum AND kinetic-energy integral equations are advanced together from the stagnation point along each surface, so the shape factor is a solved variable carrying its own history rather than a local function of the pressure gradient. The three closure functions - H*(H), Re_theta*Cf/2 and Re_theta*C_D - are properties of

the Falkner-Skan family, computed from it and not fitted; every similarity solution is an exact fixed point of the second equation, and the march returns H = 2.5914 on a flat plate against the Blasius 2.5913. Fluid properties are evaluated at Eckert's reference temperature so that the incompressible closures return the compressible skin friction.

- Unified transition kernel (the novel core): at every station the effective transition Reynolds number is the minimum across the four mechanisms, each with a calibration weight.
- Transitional region: a Narasimha universal-intermittency closure blends laminar and turbulent properties through γ.
- Turbulent boundary layer: Head's entrainment method with the Ludwieg–Tillmann skin-friction law advances the turbulent BL to the trailing edge.
- Integration: the Squire–Young formula returns profile drag; a span-wise strip sweep with the cross-flow mechanism extends the solution to the full 3-D wing.

Two elements of the formulation are not correlations. The amplification rate driving the e^N integral is tabulated in solver/amplification_db.npz from 61,600 Orr-Sommerfeld eigenvalue solutions on the Falkner-Skan family, continued past separation onto its reverse-flow branch: profiles are generated by continuation, the eigenvalue problem is solved by Chebyshev collocation, and the spatial growth rate is recovered from the temporal one through Gaster's transformation. The stability solver returns the Blasius neutral point at Re_theta = 201 against the accepted 200.5 - the tabulated database resolves it to the nearest node of its Reynolds-number grid, Re_theta = 210 - a shape factor of 2.59129 and a wall shear parameter f''(0) = 0.46960. At run time the cost is a table lookup, so the sub-second solution is preserved. The second is the separation-bubble closure described above, whose length scales with the disturbance environment and therefore spans measured bubbles from about 40 momentum thicknesses at 2.1 % free-stream turbulence to about 180 at 0.03 %.

4. The UTSS Universal Solver — Governing Equations

All equations implemented in the solver are listed below as native, editable Word equations at standard size. They constitute the complete mathematical definition of the method, and are also collected in the companion file model.equations.docx.

4.1 Inviscid edge solution

(E02) Vortex-panel surface velocity (Kuethe & Chow)

$$\frac{V_{t_i}}{U_\infty} = \cos(\theta_i - \alpha) + \sum_{j=1}^{N} C_{t,ij}\gamma_j$$

(E03) Kutta condition (sharp trailing edge)

$$\gamma_1 + \gamma_{N+1} = 0$$

(E01) Pressure coefficient (inviscid)

$$C_p = 1 - \left(\frac{V}{U_\infty}\right)^2$$

(E04) Karman-Tsien compressibility correction

$$C_p = \frac{C_{p,0}}{\beta + \left(\frac{M_\infty^2}{1+\beta}\right)\frac{C_{p,0}}{2}}, \beta = \sqrt{1 - M_\infty^2}$$

4.2 Laminar boundary layer (Thwaites)

(E05) Thwaites momentum thickness (laminar)

$$\theta^2 = \frac{0.45\nu}{U_e^6}\int_0^x U_e^5 dx'$$

(E06) Thwaites pressure-gradient and momentum-thickness Reynolds number

$$\lambda = \frac{\theta^2}{\nu}\frac{dU_e}{dx}, Re_\theta = \frac{U_e\theta}{\nu}$$

(E06b) Kinetic-energy integral (two-equation laminar march)

$$\theta\frac{dH^*}{dx} = 2C_D - H^*\frac{C_f}{2} - H^*(1-H)\frac{\theta}{U_e}\frac{dU_e}{dx}$$

(E06c) Laminar closure functions (from the Falkner-Skan family)

$$H^*(H) = \frac{\theta^*}{\theta}, l(H) = Re_\theta\frac{C_f}{2} = \theta_\eta f''(0), d(H) = Re_\theta C_D = \theta_\eta \int (f'')^2 d\eta$$

(E07) Von Karman momentum-integral equation

$$\frac{d\theta}{dx} + (2+H)\frac{\theta}{U_e}\frac{dU_e}{dx} = \frac{C_f}{2}$$

4.3 Unified four-mechanism transition kernel (novel contribution)

Bypass onset uses the Abu-Ghannam & Shaw correlation evaluated at the flow-history-averaged Tu; natural/TS onset integrates one amplification factor per physical frequency using the tabulated Orr-Sommerfeld growth rates and triggers on their envelope at N_crit; separation-induced onset closes a laminar bubble across the dead-air region; and cross-flow onset uses the C1 criterion. The kernel takes the minimum effective onset across all four:

(E08) Abu-Ghannam & Shaw bypass onset

$$Re_{\theta t} = 163 + \exp\left[F(\lambda_\theta) - \frac{F(\lambda_\theta)Tu}{6.91}\right]$$

(E09) AGS pressure-gradient function

$$F(\lambda_\theta) = \begin{cases} 6.91 + 12.75\lambda_\theta + 63.64\lambda_\theta^2, & \lambda_\theta \le 0 \\ 6.91 + 2.48\lambda_\theta - 12.27\lambda_\theta^2, & \lambda_\theta > 0 \end{cases}$$

(E10) Natural / Tollmien-Schlichting onset (e^N, envelope over frequency)

$$N(x) = \max_\omega \int_{x_0(\omega)}^x \frac{\sigma(H, Re_\theta, \omega\theta/U_e)}{\theta} dx' \ge N_{crit}$$

(E10b) Orr-Sommerfeld operator (tabulated amplification rates)

$$[(U-c)(D^2-\alpha^2)-U'']\hat{v}=\frac{1}{i\alpha Re_\theta}(D^2-\alpha^2)^2\hat{v}$$

(E10c) Gaster transformation (temporal to spatial growth rate)

$$\sigma=-\alpha_i\theta=\frac{\omega_i}{c_g},c_g=\frac{\partial\omega_r}{\partial\alpha_r}$$

(E11) Cross-flow criterion (swept wing, C1 on Re_theta2)

$$Re_{\theta 2}=k_{cf}Re_\theta\sin\Lambda\cos\Lambda\ \geq\ C_1$$

(E11b) Cross-flow amplification integral (the closure actually used)

$$N_{cf}=\int_{x_{c1}}^{x}\frac{\sigma(H_{rev})}{\theta}dx'\ \geq\ N_{crit}$$

(E12) Separation bubble: dead-air march (momentum and kinetic energy, no wall shear)

$$\frac{d\theta}{dx}=-(2+H)\frac{\theta}{U_e}\frac{dU_e}{dx},\theta\frac{dH^*}{dx}=2C_D-H^*(1-H)\frac{\theta}{U_e}\frac{dU_e}{dx},C_f=0$$

(E12b) Separation bubble: reattachment condition

$$N_{bub}=\int_{x_s}^{x_r}\frac{\sigma(H_{rev},Re_\theta)}{\theta}dx'=N_{crit},\sigma(H_{rev})\approx 0.0435$$

(E13) Unified minimum-onset transition kernel

$$Re_{\theta t}^{*}=\min\left(a_{TS}Re_{\theta t}^{TS},a_{BP}Re_{\theta t}^{BP},a_{SEP}Re_{\theta t}^{SEP},a_{CF}Re_{\theta t}^{CF}\right)$$

(E14) Transition trigger

$$Re_\theta(x_t)\geq Re_{\theta t}^{*}$$

4.4 Transitional region and turbulent closure

(E15) Narasimha universal intermittency

$$\gamma(x)=1-\exp[-0.412\xi^2],\xi=\frac{x-x_t}{\lambda_{tr}}$$

(E16) Transition-length scale

$$\lambda_{tr}=\frac{\nu}{U_e}C_{len}Re_{x,t}^{0.75},C_{len}=9$$

(E17) Intermittency-weighted property blend

$$\phi=(1-\gamma)\phi_{lam}+\gamma\phi_{turb}$$

(E18) Head entrainment (turbulent)

$$\frac{d(\theta H_1)}{dx}=C_E-\frac{\theta H_1}{U_e}\frac{dU_e}{dx},C_E=0.0306(H_1-3)^{-0.6169}$$

(E19) Ludwieg-Tillmann skin-friction law

$$C_f=0.246\times 10^{-0.678H}Re_\theta^{-0.268}$$

4.5 Drag, temperature and reference quantities

(E20) Squire-Young profile drag

$$C_d = 2\frac{\theta_{TE}}{c}\left(\frac{U_{e,TE}}{U_\infty}\right)^{(H_{TE}+5)/2}$$

(E21) Crocco-Busemann temperature profile

$$\frac{T}{T_e} = 1 + r\frac{\gamma-1}{2}M_e^2\left[1-\left(\frac{u}{U_e}\right)^2\right]$$

(E22) Recovery (adiabatic-wall) temperature

$$T_r = T_e\left(1 + r\frac{\gamma-1}{2}M_e^2\right), r \approx Pr^{1/3}$$

(E22b) Eckert reference temperature (compressible closures)

$$\frac{T_{ref}}{T_e} = 1 + 0.032M_e^2 + 0.58\left(\frac{T_w}{T_e}-1\right), \frac{\nu_{ref}}{\nu_e} = \left(\frac{T_{ref}}{T_e}\right)^{1+\omega}$$

(E23) Reynolds number (mean aerodynamic chord)

$$Re_{MAC} = \frac{\rho_\infty U_\infty \bar{c}}{\mu_\infty} = \frac{U_\infty \bar{c}}{\nu_\infty}$$

(E24) Mean aerodynamic chord

$$\bar{c} = \frac{2}{3}c_{root}\frac{1+\lambda+\lambda^2}{1+\lambda}$$

5. Why UTSS is Better — Comparison with Existing Solvers

UTSS is designed to be fast, robust and broad in regime coverage. The table contrasts its capabilities with the principal classes of existing transition tools.

Capability	XFOIL / e^N codes	RANS γ–Re_θ (CFD)	LES / DNS	UTSS (this work)
Natural / TS transition	Yes	Indirect	Yes	Yes (tabulated e^N)
Bypass (free-stream Tu)	No	Yes	Yes	Yes (AGS)
Separation-induced	Limited	Yes	Yes	Yes (bubble closure)
Cross-flow (3-D swept)	No	Add-on only	Yes	Yes (C1, built-in)
Single universal calibration	n/a	Needs re-tuning	n/a	Yes (one constant set)
3-D wing capability	2-D only	Yes	Yes	Yes (strip + cross-flow)
Robustness / convergence	Can fail in sep.	Stiff, costly	Very costly	Robust, direct
Typical CPU cost	seconds	hours	days–weeks	< 1 second
Suited to design loops	Partly	No	No	Yes

Table 2. Capability comparison versus existing solver classes.

Novelty. The distinguishing element is the unified transition kernel (Eq. E13): a single closed expression that selects the governing transition mechanism locally by taking the minimum effective onset Reynolds number across four co-resident mechanisms, each modulated by a calibration weight, and feeds a single intermittency closure. Unlike e^N codes (TS only) or correlation RANS models (which require case-by-case re-tuning and a full CFD solve), UTSS reproduces natural, bypass, separation and cross-flow transition with ONE calibration set at panel-method cost. Every one of the four branches is the selected mechanism somewhere in this study — natural/TS on the cruise wing, the Schubauer-Skramstad plate and the NLF(1)-0416 upper surface; bypass on the climb case and the ERCOFTAC plates; separation on T3C4 and the NLF(1)-0416 lower surface; cross-flow on both swept wings — and each is checked against measurement in Section 11.

6. Case-Study Definition and Input Data

All input data required to run the prediction are tabulated below. The case is fully three-dimensional.

6.1 Aircraft and wing geometry (input)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m^2
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section c_l at cruise design incidence (alpha = 1.5 deg, M = 0.42)	0.517	-

Table 3. Geometry definition (input).

6.2 Flight conditions (input)

case	altitude_m	mach	U_inf_ms	rho_kgm3	mu_Pas	T_inf_K
CRUISE (FL360, M0.42)	11000.0	0.42	123.93	0.3639	1.422e-05	216.65
CLIMB (FL100, M0.30)	3000.0	0.3	98.57	0.9093	1.694e-05	268.66

(columns 2-7 of 12; case repeated on each block)

case	Tu_pct	alpha_deg	Re_MAC	nu_m2s	q_inf_Pa
CRUISE (FL360,	0.07	1.5	6414000.0	3.907e-05	2794.6

M0.42)					
CLIMB (FL100, M0.30)	0.9	3.5	10700000.0	1.863e-05	4417.8

Table 4. Flight / flow conditions (input). (columns 8-12 of 12; the table is split across 2 blocks so that no cell is compressed to illegibility, with case repeated on each)

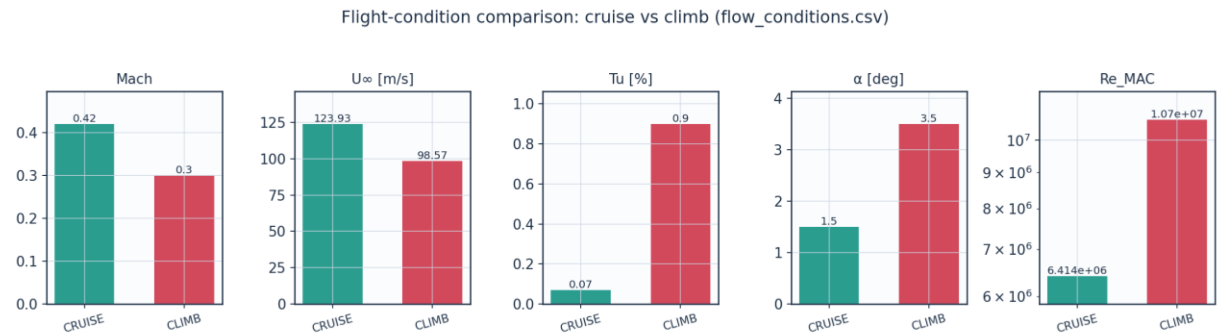


Fig. 8a. Cruise vs climb flight-condition comparison (flow_conditions.csv).

6.3 Fluid (material) properties (input)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m^3

Table 5. Air properties (input).

6.4 Solver settings (input)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility (pressure)	Karman-Tsien correction on C_p
Compressibility (boundary layer)	closures evaluated at Eckert's reference temperature
Laminar BL closure	two-equation march (momentum + kinetic energy); H*, l and d from the Falkner-Skan family
Transition model	UTSS unified 4-mechanism kernel
TS / natural	e^N, one factor per physical frequency, growth rates from the tabulated Orr-Sommerfeld database
bypass	Abu-Ghannam & Shaw at the flow-history-averaged Tu
separation	laminar bubble closed by the amplification integral across the dead-air region
cross-flow	C1 on Re_theta2, closed by an amplification integral
Intermittency closure	Narasimha universal (gamma)
Transition length	Dhawan & Narasimha, Re_lambda = 9 Re_x_t^0.75
Turbulent BL closure	Head entrainment + Ludwieg-Tillmann Cf
Laminar separation criterion	Thwaites lambda <= -0.09
Turbulent separation flag	H > 2.6

Drag integration	Squire-Young far-wake, evaluated at 0.98c
Marching scheme	explicit trapezoidal, arc-length stepping with sub-stepping on the kinetic-energy equation
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	~1 (reconstruction grid)

Table 6. Solver configuration (input).

6.5 Universal calibration constants (input)

constant	value	role	calibration_source
A_TS	1.0	TS/natural onset weight	unity; validated on Schubauer-Skramstad and NLF(1)-0416 upper surface
A_BP	1.0	bypass onset weight	unity; validated on ERCOFTAC T3A/T3A-/T3B
A_SEP	1.0	separation-induced weight	unity; validated on T3C4 and NLF(1)-0416 lower surface
A_CF	1.0	crossflow weight	unity; Arnal C1 criterion
N_crit	from Tu	e^N amplification factor	Mack N=-8.43-2.4 ln(Tu), clamped to 0.0008<=Tu<=0.0298, anchored -0.50 (FITTED jointly on S&S and 86 aerofoil pts); 8.18 for Tu<0.08%
N_floor	0.5	lower clamp on N_crit at high Tu	clamp; bypass governs there
Tu_TS_max	0.1	upper Tu for the attached-flow TS branch	complement of the bypass gate
C_len	9.0	transition-length scale, Re_lambda = C_len Re_x_t^0.75	Dhawan & Narasimha (1958) published value; not fitted here
CF_C1	150.0	crossflow critical Re_theta2	Arnal et al.; facility-dependent: 150 fits Dagenhart (14.7%), 200 fits Boltz (18.4%)
cf_amp	True	crossflow closed by an amplification integral, not a local threshold	rate computed (0.0435); threshold is the same N_crit; adds no constant
CF_ratio	0.47	theta2/theta surrogate for crossflow	FITTED on Dagenhart & Saric; independent check in Sec. IV.C
sigma_sep	from table	shear-layer amplification rate in a bubble	computed on the reverse-flow Falkner-Skan branch; 0.0435, not fitted
two_eq	True	two-equation laminar march (momentum + kinetic energy)	closures from the Falkner-Skan family; no fitted constant
N_anchor	0.5	offset on Mack's N_crit	unit conversion to the present amplification rates; set jointly on S&S and the 86 NLF(1)-0416 conditions
C_mu	0.09	k-epsilon constant in the free-stream turbulence decay	standard value 0.09; with C_eps2 it fixes Tu(x) from the rig's integral length scale, and neither is fitted here
C_eps2	1.92	k-epsilon constant in the free-stream turbulence decay	standard value 1.92; sets the decay exponent of isotropic grid turbulence
bubble	True	separation branch closed by marching the detached shear layer	True is the model; False is the 'no bubble closure' ablation
use_os_db	True	amplification rates read from the tabulated Orr-Sommerfeld solutions	True is the model; False selects the Drela-Giles envelope fit, the third ablation
cf_exact	False	exact Falkner-Skan-Cooke K(lambda) in	False is the model; True was tried and is reported in Sec. IV.C, where it makes the two

		place of the constant surrogate (ablation path only)	swept wings agree less, not more
H_sep	0.0	declare laminar separation on the solved shape factor instead of on lambda (ablation path only)	0 = use lam_sep; the shape-factor test is the more principled statement but the aerofoil measurements reject it - see the note in march_bl
tu_hist	0.6	weight on the flow-history average of Tu, Tu_eff = Tu_avg^w Tu_local^(1-w)	FITTED on the three ERCOFTAC plates; no effect where Tu does not decay
lam_sep	-0.09	Thwaites parameter at laminar separation	Thwaites (1949) published value; not fitted
sep_floor	120.0	min separation-induced onset Re_theta (ablation path only)	superseded by the bubble closure; reachable only with bubble=False
n_freq	40	frequencies carried by the e^N integration	discretisation parameter; onset moves <0.2% over 24-64
CF_N	0.0	separate amplification threshold for the crossflow branch (ablation path only)	0 = use the same N_crit as every other branch; exposed to test whether a roughness-seeded branch needs its own threshold, and it does not - see 06_validation/crossflow_criticals_summary.csv

Table 7. Calibration constants and their sources (input).

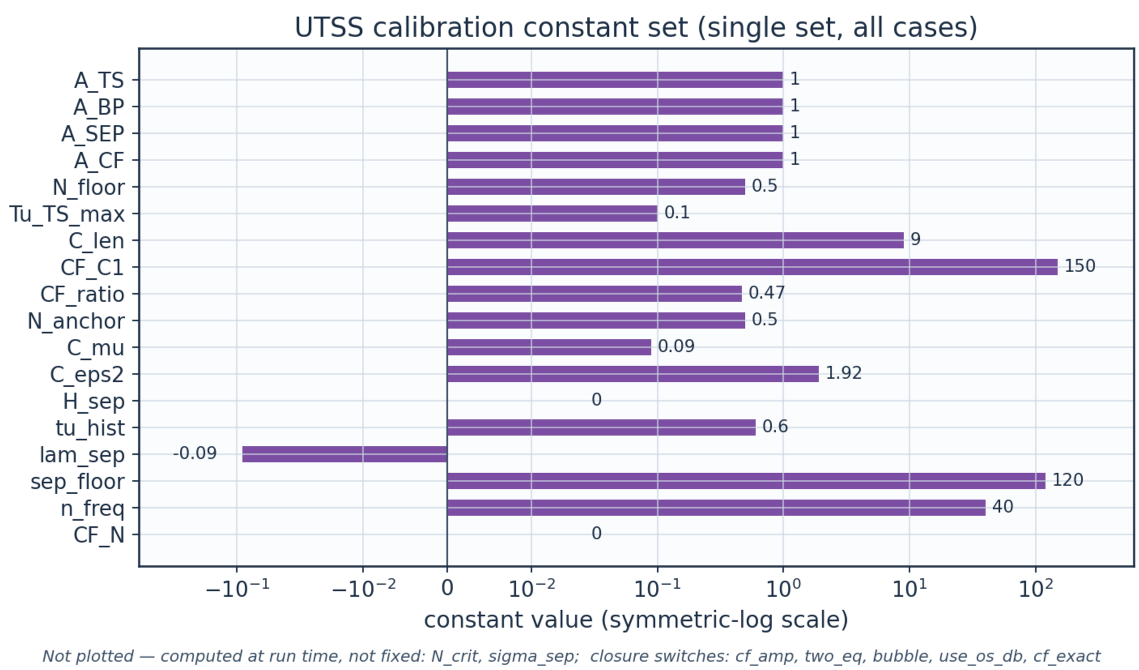


Fig. 8b. UTSS universal calibration constant set (calibration_constants.csv).

7. Geometry and Engineering Drawings

The wing is drawn to standard third-angle orthographic projection. All drawings are dimensioned and to scale; views provided: section, plan, front, side, full orthographic sheet, isometric and structural section.

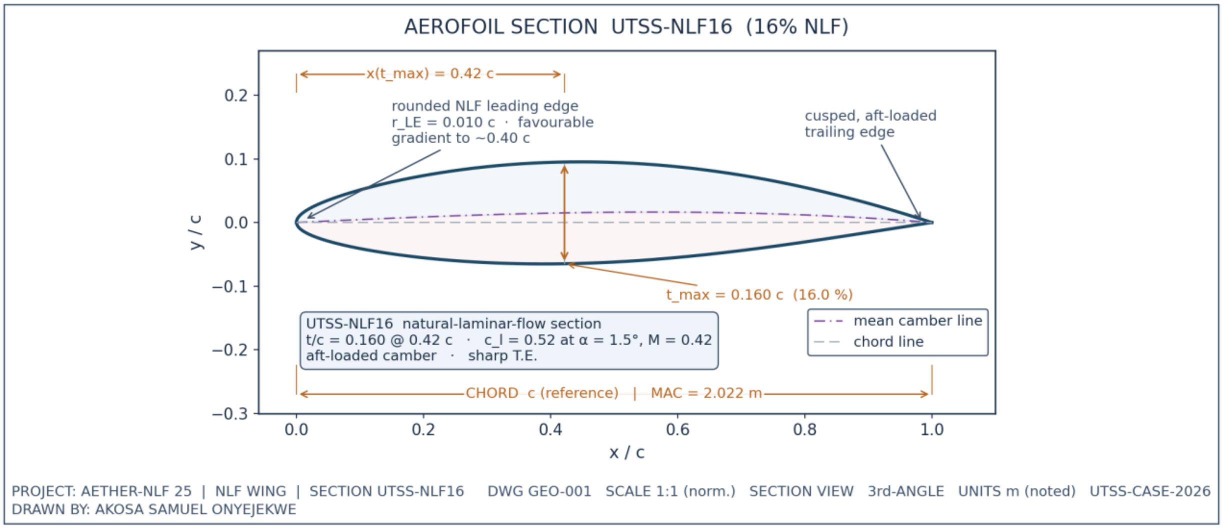


Fig. 1. UTSS-NLF16 aerofoil section (dimensioned).

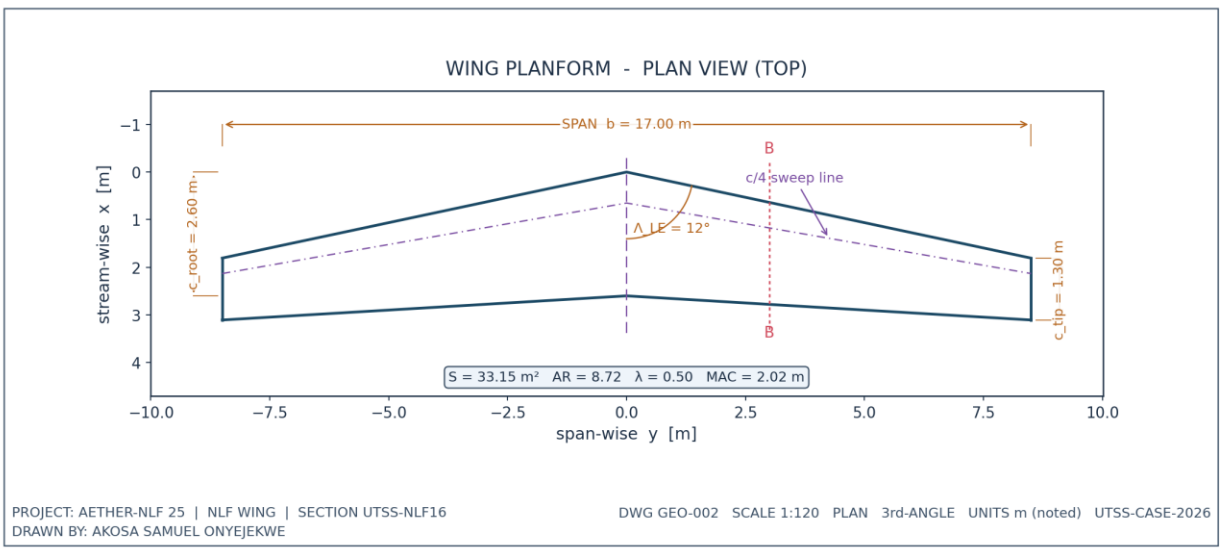


Fig. 2. Wing planform — plan (top) view, dimensioned.

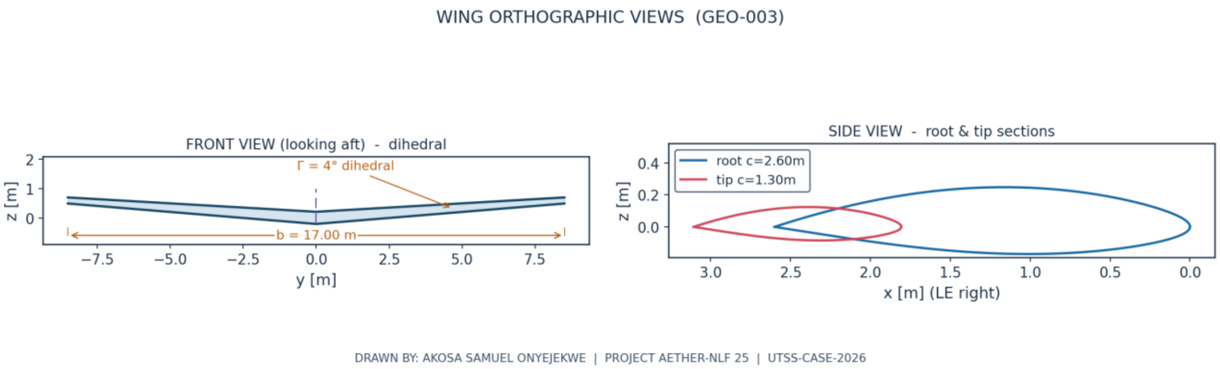


Fig. 3. Front and side orthographic views.

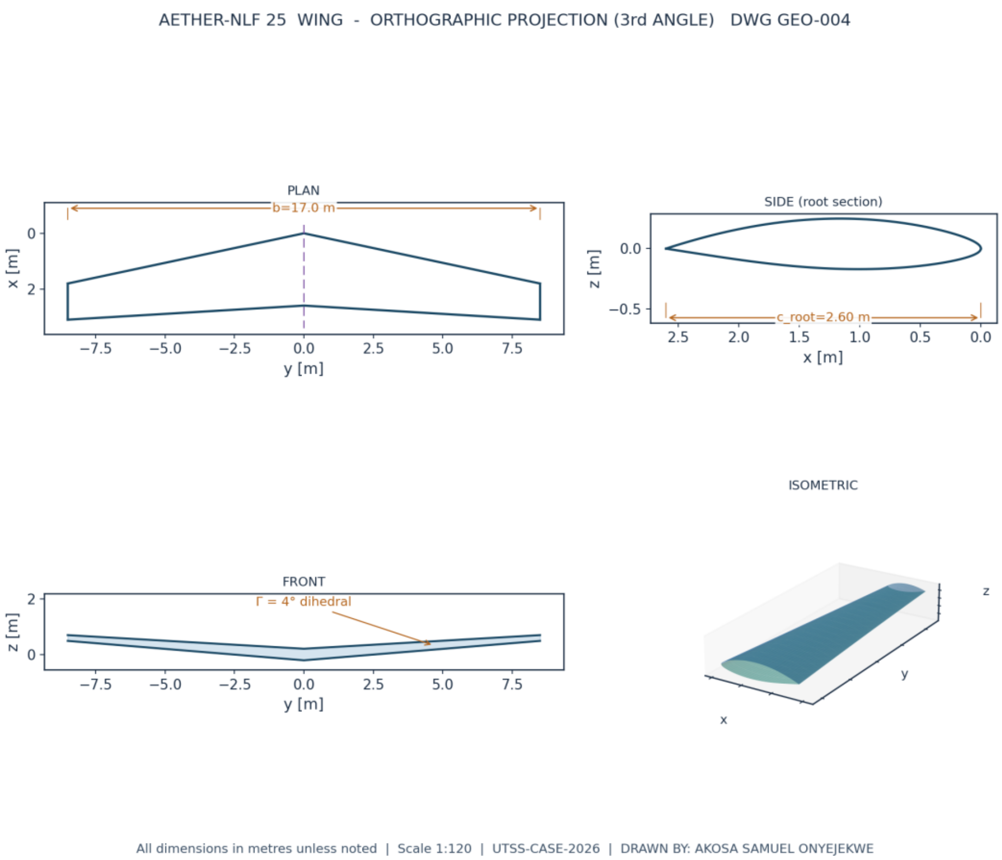


Fig. 4. Full third-angle orthographic projection sheet.

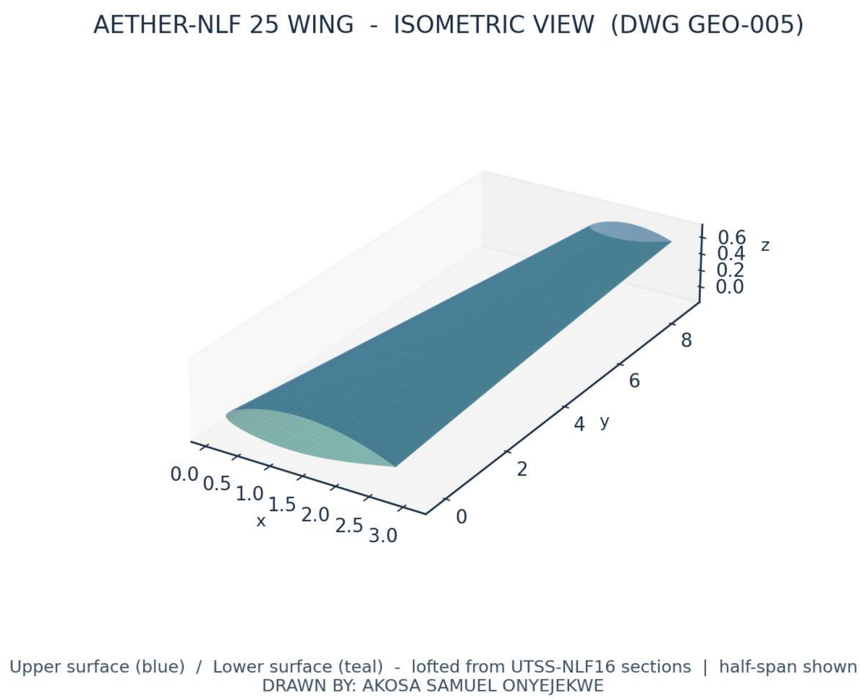


Fig. 5. Isometric view of the wing.

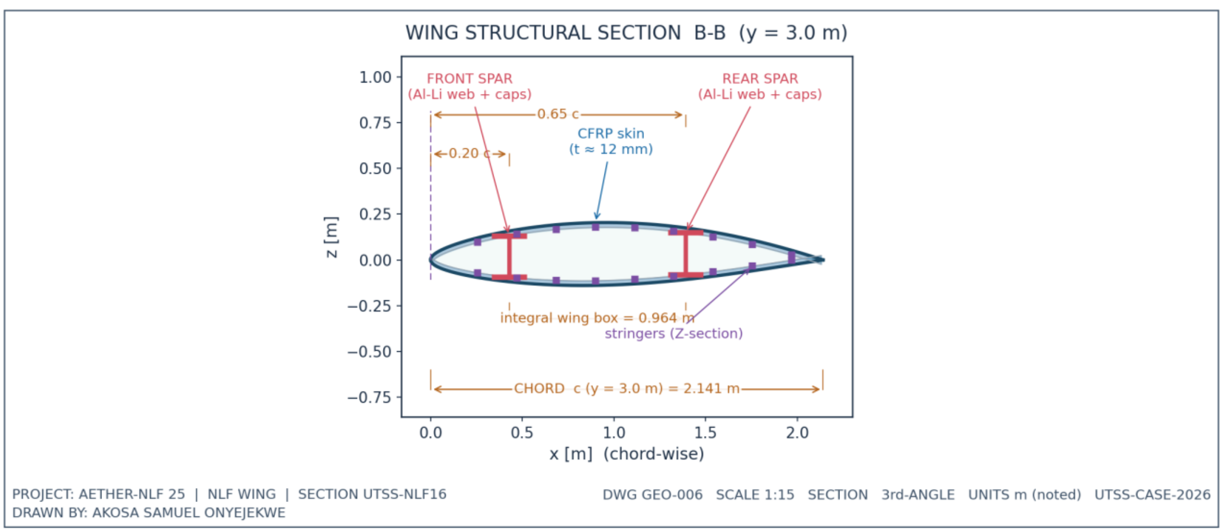


Fig. 6. Structural sectional view B–B.

7.1 Geometry data (from CSV)

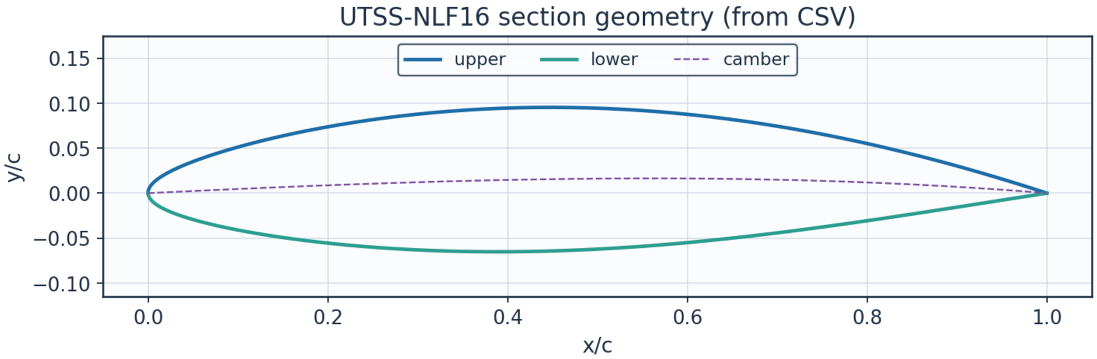


Fig. 7. Section geometry plotted from airfoil_UTSS-NLF16.csv.

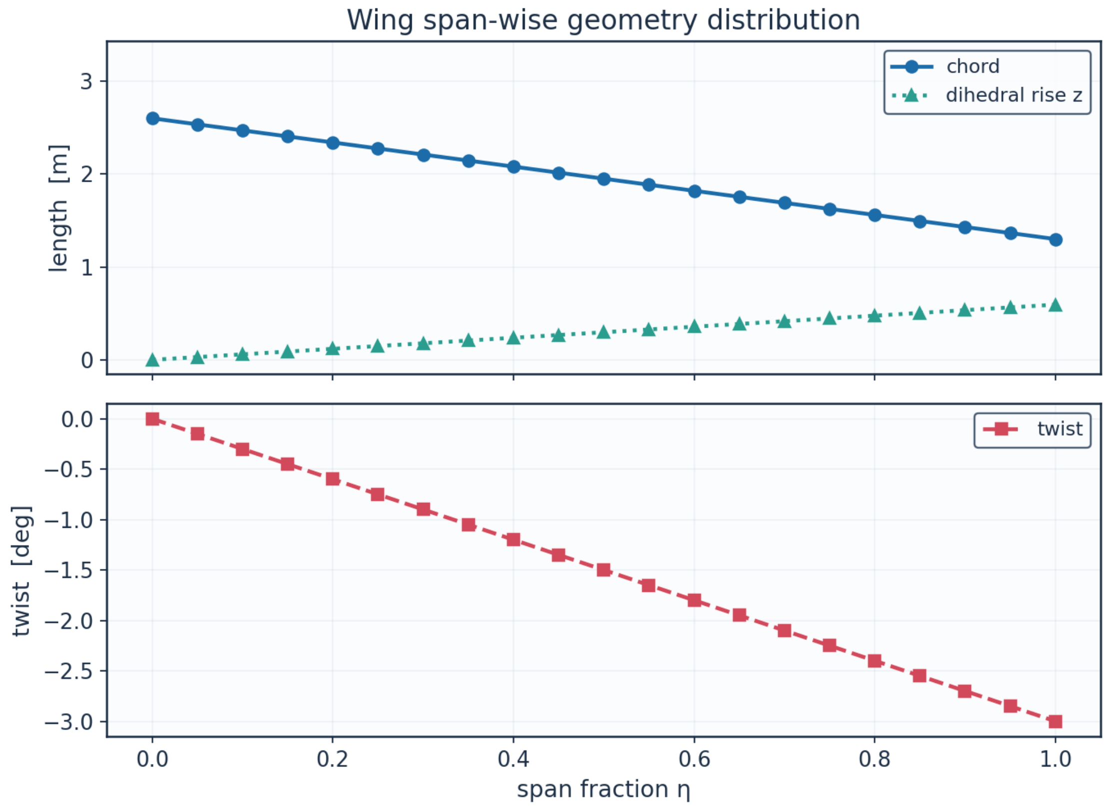


Fig. 8. Span-wise geometry from wing_planform.csv.

eta	y_m	chord_m	x_le_m	x_te_m	twist_deg	z_dihedral_m
0.0	0.0	2.6	0.0	2.6	-0.0	0.0
0.05	0.42500000000000004	2.535	0.09033653870975941	2.6253365387097594	-0.15000000000000002	0.02971889507599193
0.1	0.8500000000000001	2.47	0.18067307741951882	2.650673077419519	-0.30000000000000004	0.05943779015198386
0.15000000000000002	1.2750000000000001	2.4050000000000002	0.27100961612927826	2.6760096161292783	-0.45000000000	0.08915668522797579

					000007	
0.2	1.7000000000000002	2.34	0.36134615483903765	2.7013461548390376	-0.6000000000000001	0.11887558030396772
0.25	2.125	2.275	0.45168269354879703	2.726682693548797	-0.75	0.14859447537995962
0.30000000000000004	2.5500000000000003	2.21	0.5420192322585565	2.7520192322585566	-0.9000000000000001	0.17831337045595158
0.35000000000000003	2.975	2.145	0.6323557709683159	2.777355770968316	-1.05	0.20803226553194348
0.4	3.4000000000000004	2.08	0.7226923096780753	2.8026923096780756	-1.2000000000000002	0.23775116060793544
0.45	3.825	2.015	0.8130288483878346	2.828028848387835	-1.35	0.26747005568392734
0.5	4.25	1.9500000000000002	0.9033653870975941	2.853365387097594	-1.5	0.29718895075991925
0.55	4.675000000000001	1.885	0.9937019258073536	2.878701925807354	-1.6500000000000001	0.32690784583591126
0.6000000000000001	5.1000000000000005	1.8199999999999998	1.084038464517113	2.9040384645171127	-1.8000000000000003	0.35662674091190316
0.65	5.525	1.755	1.1743750032268723	2.929375003226872 4	-1.9500000000000002	0.38634563598789506
0.7000000000000001	5.95	1.69	1.2647115419366317	2.9547115419366317	-2.1	0.41606453106388697
0.75	6.375	1.625	1.3550480806463912	2.980048080646391	-2.25	0.44578342613987887
0.8	6.800000000000001	1.56	1.4453846193561506	3.0053846193561506	-2.4000000000000004	0.4755023212158709
0.8500000000000001	7.2250000000000005	1.4949999999999999	1.535721158065 91	3.03072115806591	-2.5500000000000003	0.5052212162918628
0.9	7.65	1.43	1.6260576967756692	3.056057696775669	-2.7	0.5349401113678547
0.9500000000000001	8.075000000000001	1.365	1.716394235485429	3.081394235485429	-2.85	0.5646590064438467
1.0	8.5	1.3	1.8067307741951881	3.106730774195188	-3.0	0.5943779015198385

(columns 2-7 of 8; eta repeated on each block)

eta	Re_local
0.0	8246203.012995781
0.05	8040047.9376708865
0.1	7833892.862345993
0.15000000000000002	7627737.787021098
0.2	7421582.711696202
0.25	7215427.636371308
0.30000000000000004	7009272.561046414
0.35000000000000003	6803117.485721519
0.4	6596962.410396625
0.45	6390807.33507173
0.5	6184652.2597468365
0.55	5978497.184421942
0.6000000000000001	5772342.109097046

0.65	5566187.033772152
0.7000000000000001	5360031.958447257
0.75	5153876.883122363
0.8	4947721.807797468
0.8500000000000001	4741566.732472573
0.9	4535411.6571476795
0.9500000000000001	4329256.581822785
1.0	4123101.5064978907

Table 8. Wing planform stations (wing_planform.csv). (columns 8-8 of 8; the table is split across 2 blocks so that no cell is compressed to illegibility, with eta repeated on each)

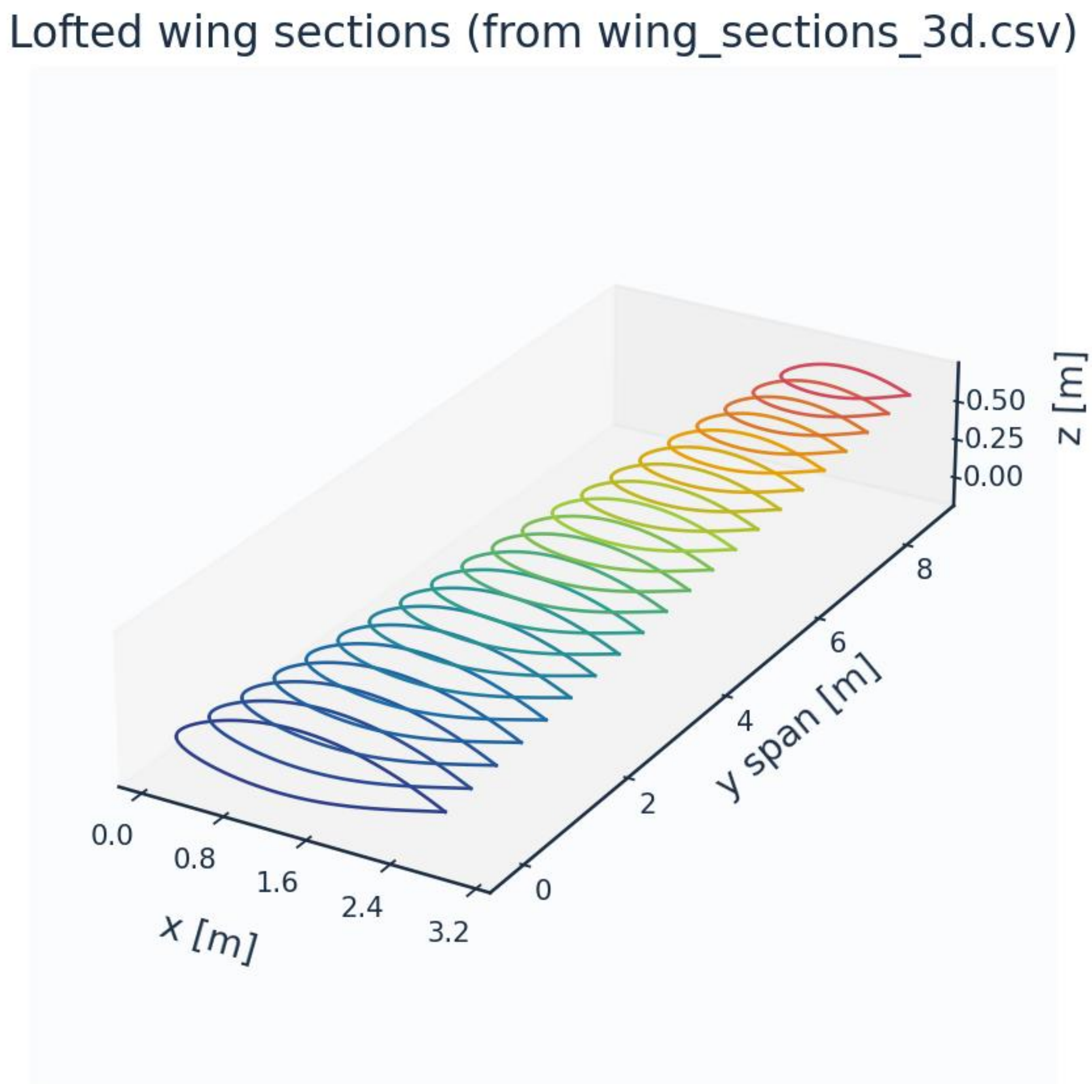


Fig. 8c. Lofted 3-D wing sections (wing_sections_3d.csv).

8. Mesh / Discretisation

The surface is discretised with cosine-clustered streamwise nodes; a wall-normal reconstruction grid (first-cell $y^+ \approx 1$) supports boundary-layer profile recovery. A panel-count sweep bounds the discretisation sensitivity rather than demonstrating asymptotic convergence: from 120 to 480 surface panels the section drag spans 2.1 counts, 4.8 % of its mean, and above 180 panels it stays within about ±0.4 count without tightening further. The residual wander is not a truncation error that refinement removes — it is set by which panel the transition point lands on, so it scales with the panel spacing at transition and is of the same order as the ±0.025c bracket the aerofoil measurements themselves carry. The 260-panel grid is used for every case-study result; the tabulated validation sections are re-splined onto their own cosine-clustered grids of 400 and 440 panels.

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y1	6.51	micron
Target wall y+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	1e-3 c
Max streamwise spacing	9.927	1e-3 c
Max streamwise spacing at MAC	20.074	mm
LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_tau (TE)	4.800	m/s
Max cell aspect ratio (at MAC)	3082	-
Discretisation type	surface panel distribution + wall-normal reconstruction stack (no volume mesh is generated)	-

Table 9. Mesh metrics.

n_surface_panels	Cl	Cd	x_tr_upper_c	dCd_pct
120	0.5146538348994995	0.004246488416072312	0.538986194980472	
180	0.5161840825332155	0.0043674049924125515	0.5433865690013859	2.84744863267703
260	0.5170864702757358	0.004373785574805958	0.5541908676652394	0.1460955053284696
360	0.5176435398599823	0.004454493992254526	0.5477820283105952	1.845276044474331
480	0.5180064091537756	0.004426592194530746	0.5554577472375675	-0.6263741240261189

Table 10. Panel-count sensitivity study.

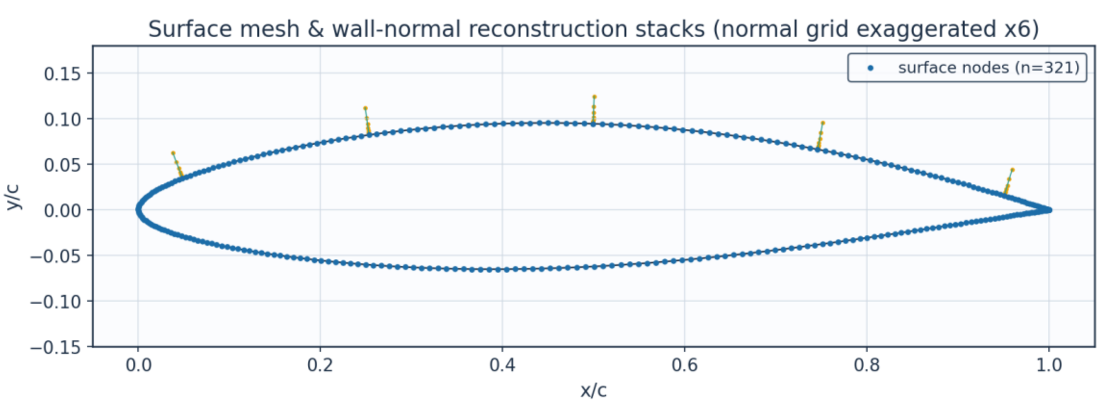


Fig. 9. Surface mesh and wall-normal stacks.

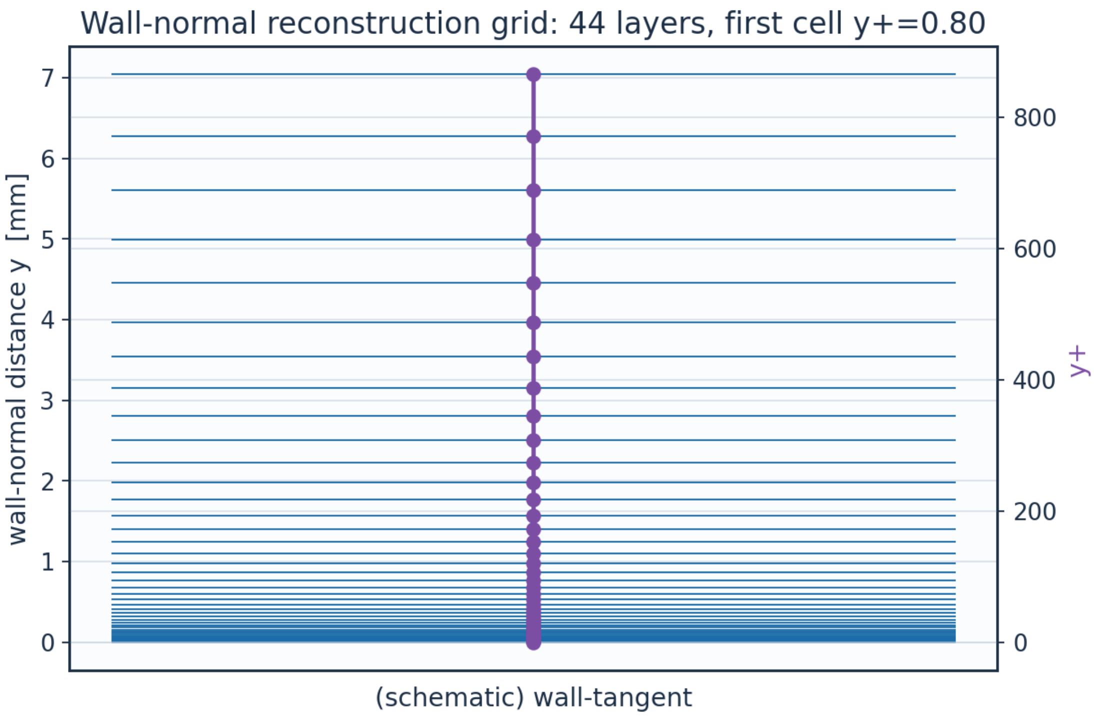


Fig. 10. Wall-normal reconstruction grid / y^{+}.

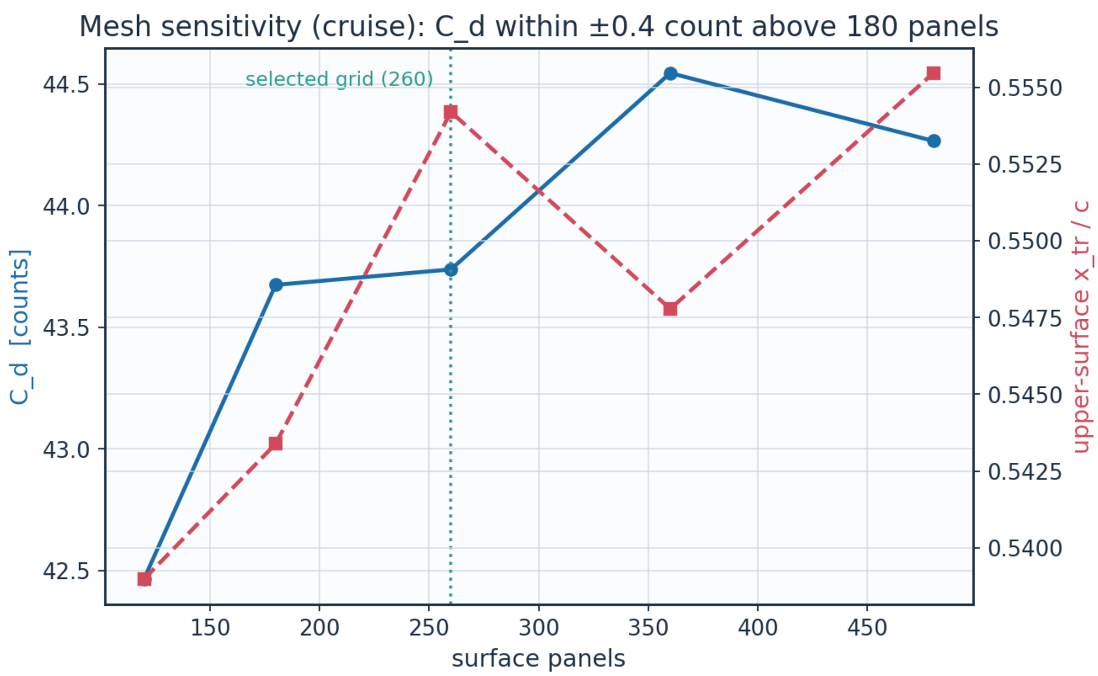


Fig. 11. Panel-count sensitivity of C_d and transition location.

layer	y_m	y_plus	cell_dy_m	growth_ratio
0	0.0	0.0	6.5127391337287455e-06	1.0
2	1.3807006963504941e-05	1.696	7.731923899562769e-06	1.12

4	3.112651649852554e-05	3.8234624	9.698925339611536e-06	1.1199999999999997
6	5.2852109259255385e-05	6.4921512345600005	1.2166331946008717e-05	1.12
8	8.010469281831492e-05	9.839754508632067	1.526144679307334e-05	1.1199999999999997
10	0.0001142903336347992	14.038988055628069	1.9143958857231194e-05	1.1199999999999997
12	0.00015717280147499707	19.30650661697985	2.4014181990510814e-05	1.1199999999999999
14	0.0002109645691337413	25.914081900339532	3.012338988889677e-05	1.12
16	0.0002784409624848701	34.20262433578592	3.778678027663213e-05	1.120000000000001
18	0.00036308335030452605	44.599771966809854	4.739973717900734e-05	1.1199999999999997
20	0.0004692587615855025	57.64195395516631	5.945823031734682e-05	1.1199999999999999
22	0.0006024451974963594	74.00206704136062	7.458440411007986e-05	1.12
24	0.0007695142627029383	94.52419289668278	9.355867651568417e-05	1.1199999999999999
26	0.0009790856980980709	120.26714756959889	0.00011736000382127424	1.1200000000000006
28	0.0012419721066577252	152.55910991130486	0.00014721638879340647	1.1200000000000008
30	0.0015717368175549556	193.06614747274082	0.00018466823810244903	1.1199999999999997
32	0.0019853936709044416	243.87817538980616	0.00023164783787571225	1.1200000000000003
34	0.002504284827746037	307.6167832089729	0.0002905790478312934	1.119999999999999
36	0.0031551818948881345	387.57049285733575	0.0003645023575995746	1.12
38	0.003971667175911181	487.864426240242	0.00045723175737290614	1.1199999999999999
40	0.004995866312426492	613.6731362757596	0.0005735515164485738	1.1200000000000014
42	0.006280621709271296	771.4875821443129	0.0007194630222330908	1.1199999999999997

Table 11. Wall-normal grid (bl_normal_grid.csv, sampled).

(table sampled to 22 rows; full data in 02_mesh/bl_normal_grid.csv)

9. Solution — Generated Engineering Output Data

This section presents every generated output: numerical tables (CSV) and the plotted curve for each. The boundary-layer state is reported on both surfaces at the cruise and climb conditions.

9.1 Transition prediction summary

case	surface	s_tr_c	x_tr_c	Re_x_tr	Re_theta_at_ onset	mechanism
CRUISE	upper	0.579	0.554	3.712e+06	1410.4	TS-natural
CRUISE	lower	0.587	0.578	3.766e+06	1270.7	TS-natural
CLIMB	upper	0.047	0.025	5.069e+05	483.8	bypass
CLIMB	lower	0.159	0.157	1.706e+06	610.7	bypass

(columns 2-7 of 9; case repeated on each block)

case	laminar_run_pct	Cf_te
CRUISE	55.4	0.00021
CRUISE	57.8	0.00025
CLIMB	2.5	0.00014
CLIMB	15.7	0.00019

Table 12. Transition prediction summary (transition_summary.csv). (columns 8-9 of 9; the table is split across 2 blocks so that no cell is compressed to illegibility, with case repeated on each)

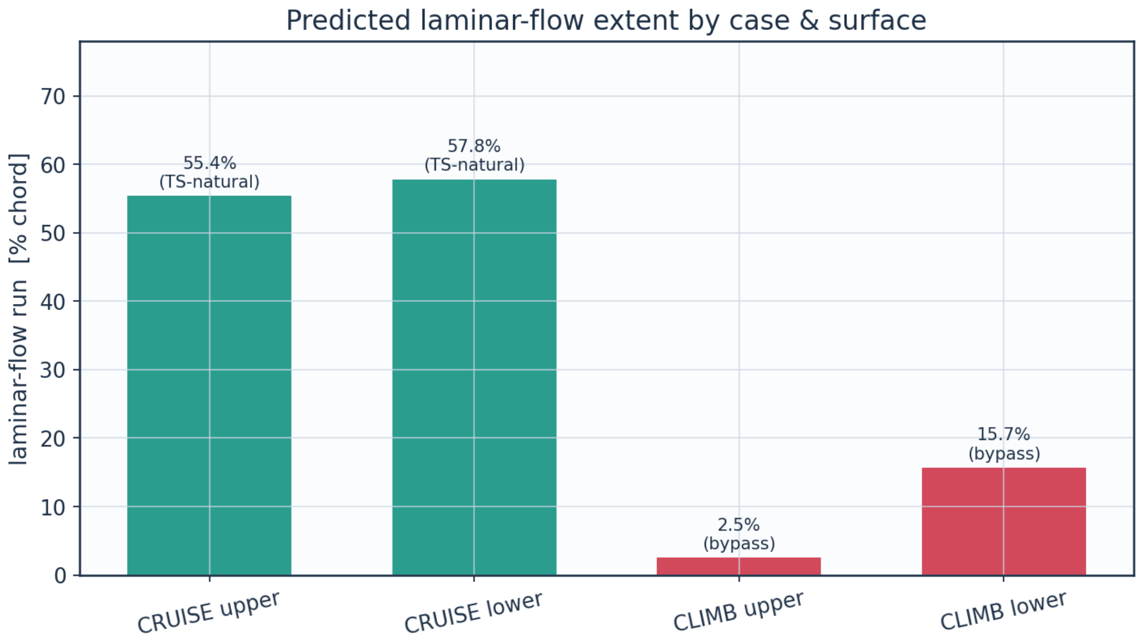


Fig. 12a. Predicted laminar-flow extent by case and surface (transition_summary.csv).

9.2 Integrated forces and drag breakdown

The wing lift is obtained from Prandtl's lifting-line theory - Glauert's monoplane equation solved for the odd Fourier coefficients of the loading - using the section lift-curve slope and zero-lift incidence returned by the same panel method, the planform chord distribution, the wing's washout, and the section slope reduced by the cosine of the quarter-chord sweep. An earlier version of this table asserted C_L = 0.90 c_l; the planform actually returns 0.49, because a −3° washout is large relative to the 3.7° by which the root exceeds its zero-lift incidence, and because the downwash of an AR = 8.7 wing reduces the slope by a quarter. The span efficiency and the induced drag come out of the same solve. The implementation is checked against the one case with a closed-form answer - an elliptic planform, for which it returns e = 1 and the exact lift-curve slope to machine precision - every time the solution is regenerated.

The incidence in Table 4 is the SECTION design incidence, not the trim point of the aircraft the planform belongs to: at 1.5° the wing carries 23.6 kN against an all-up weight of 83.4 kN, and level flight at MTOW would need 7.6°. Table 13 reports both, so the two are not confused. Every transition result in this study is quoted at the design incidence and is unaffected by that distinction.

quantity	value	unit
Section lift coefficient Cl	0.5171	-
Section profile drag Cd	0.00437	-
Section Cd (counts)	43.7	counts
Section L/D	118.2	-
Section lift-curve slope a0 (panel)	8.111	1/rad
Zero-lift incidence a_L0 (panel)	-2.185	deg
Quarter-chord sweep	9.89	deg
Wing C_L (lifting line, taper + washout + sweep)	0.2548	-

Wing C_L / section c_l	0.493	-
Span efficiency e (lifting line)	0.8718	-
Wing induced drag C_Di (lifting line)	0.00272	-
Wing lift (lifting line)	23603.0	N
Wing C_L for level flight at MTOW	0.8998	-
Incidence for that C_L (lifting line)	7.55	deg
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6.414e+06	-
Mach number	0.42	-

Table 13. Integrated forces, with the wing quantities from the lifting-line solve.

configuration	Cd_counts	mean_laminar_pct	viscous_drag_reduction_pct
NLF (UTSS predicted transition)	43.7	56.6	50.5
Fully turbulent (LE trip)	88.4	0.0	0.0

Table 14. NLF vs fully-turbulent drag.

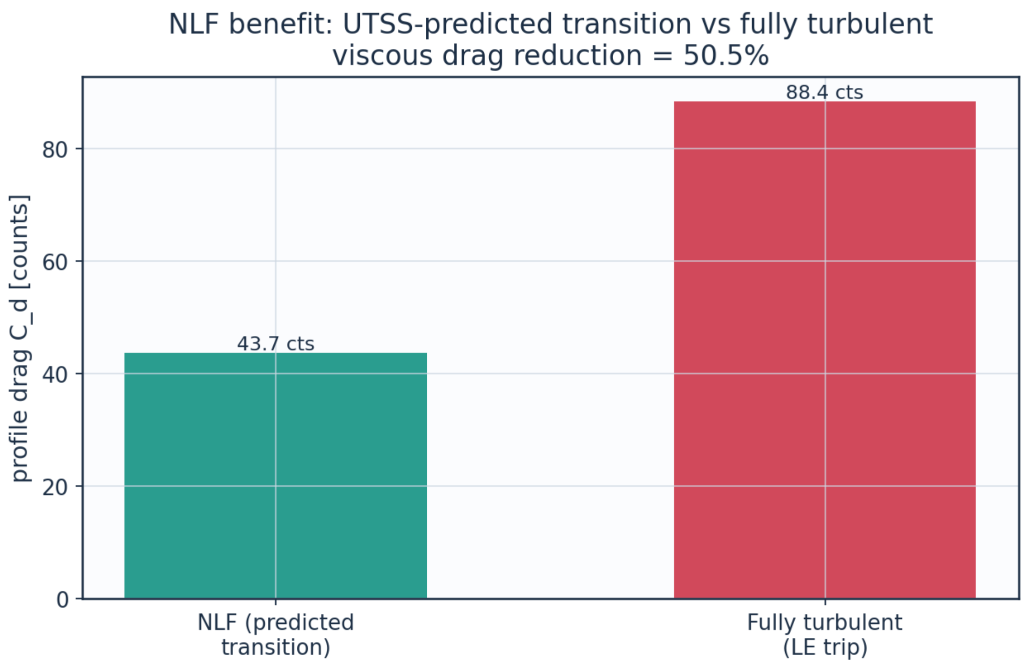


Fig. 12. Drag benefit of predicted laminar flow vs fully-turbulent.

9.2a The transition-length closure and what the result owes to it

The extent of the transitional region is Dhawan and Narasimha's published correlation, Re_λ = 9 Re_x,t^0.75, with λ the distance over which the intermittency rises from 0.25 to 0.75. It is validated here on the flat plates of Section 11.1, which span Re_x,t = 6×10^4 to 1.4×10^6 and on which it reproduces the measured extent of the skin-friction rise to within a factor of two — exactly on T3B. An earlier version of this work wrote the correlation as 6.5 Re_θt^0.8, which returns Re_λ ≈ 600 where those plates require ≈ 4×10^4: the layer went from fully laminar to fully turbulent inside a single marching station, and the predicted C_f jumped vertically where the measurements climb over half a decade of Re_x. The transition LOCATION was never affected —

onset is where the kernel fires, not where the blend ends — but everything downstream of onset was.

The cruise wing transitions at Re_x,t = 3.7×10^6, a factor of three beyond the range the correlation is validated over here, and it then returns a transitional zone of about a third of the chord, which is longer than a real natural-laminar-flow section shows at this Reynolds number. That extrapolation is not damped: doing so would add an undeclared constant to a method whose claim is that it has none. What it costs is measured instead. Sweeping the constant over a factor of four moves the section drag by a tenth of a count, so no reported drag in this study depends on the extrapolated part of the closure; beyond twice the published value the layer no longer completes transition before the trailing edge, and there it would.

C_len	multiple_of_published	Cd_counts	x_tr_c_upper	transitional_extent_pct_chord	completes_before_TE
2.25	0.25	43.84	0.554	8.3	True
4.5	0.5	43.84	0.554	18.3	True
9.0	1.0	43.74	0.554	36.3	True
18.0	2.0	39.45	0.554		False
36.0	4.0	32.84	0.554		False

Table 14a. Sensitivity of the case-study result to the transition-length constant (transition_length_sensitivity.csv).

9.3 Surface distributions — cruise

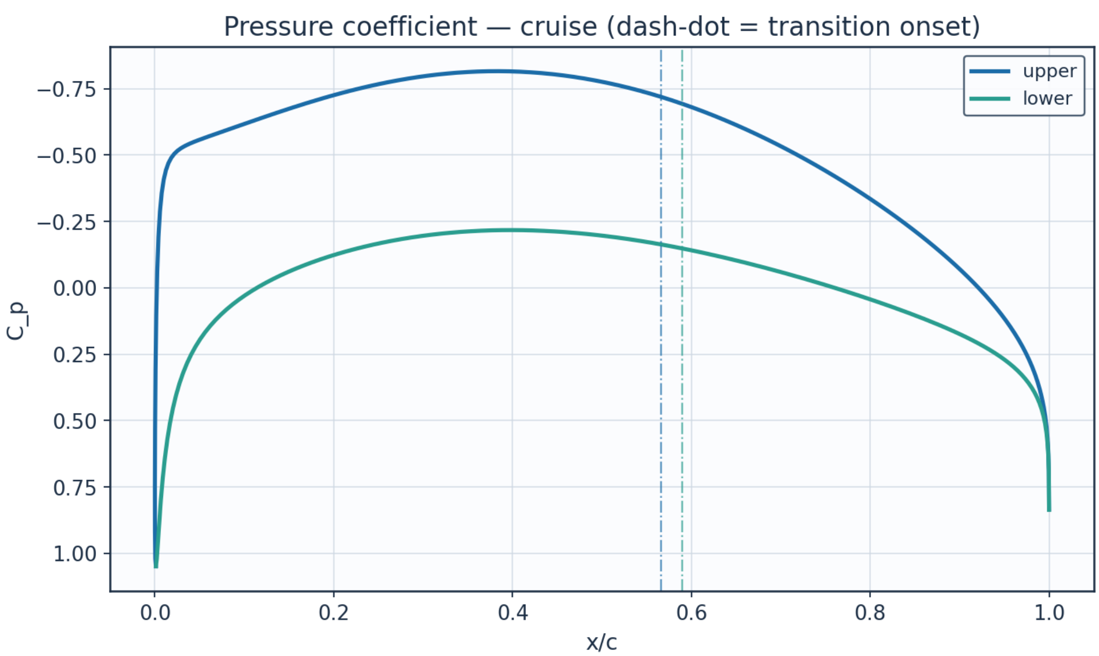


Fig. 13. Pressure coefficient C_p — cruise.

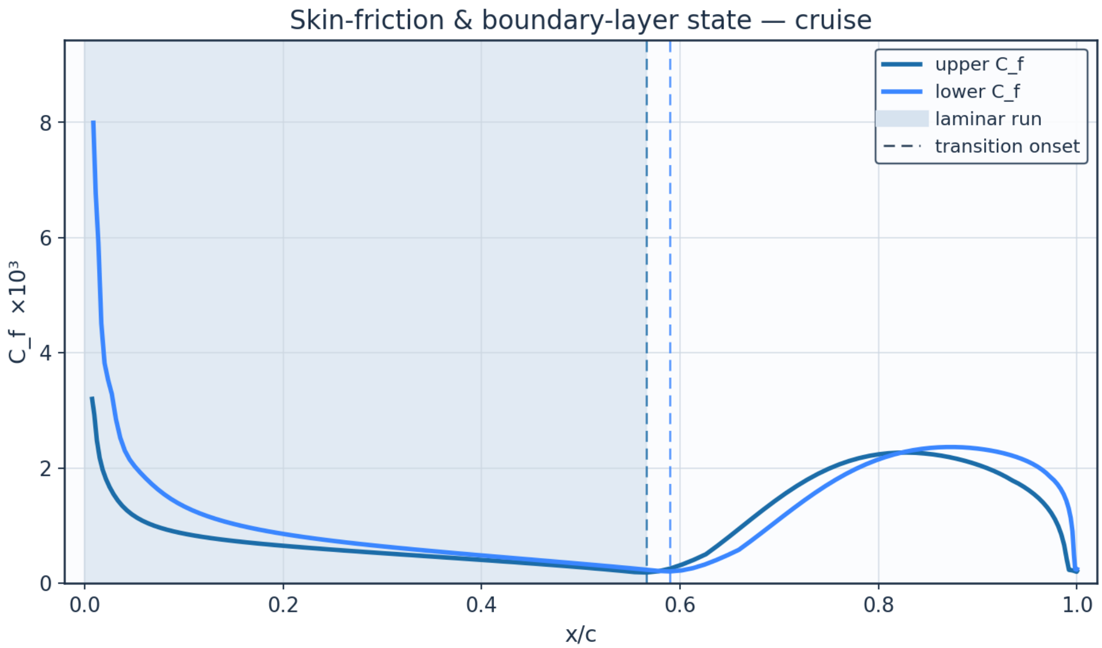


Fig. 14. Skin-friction C_f and laminar run — cruise.

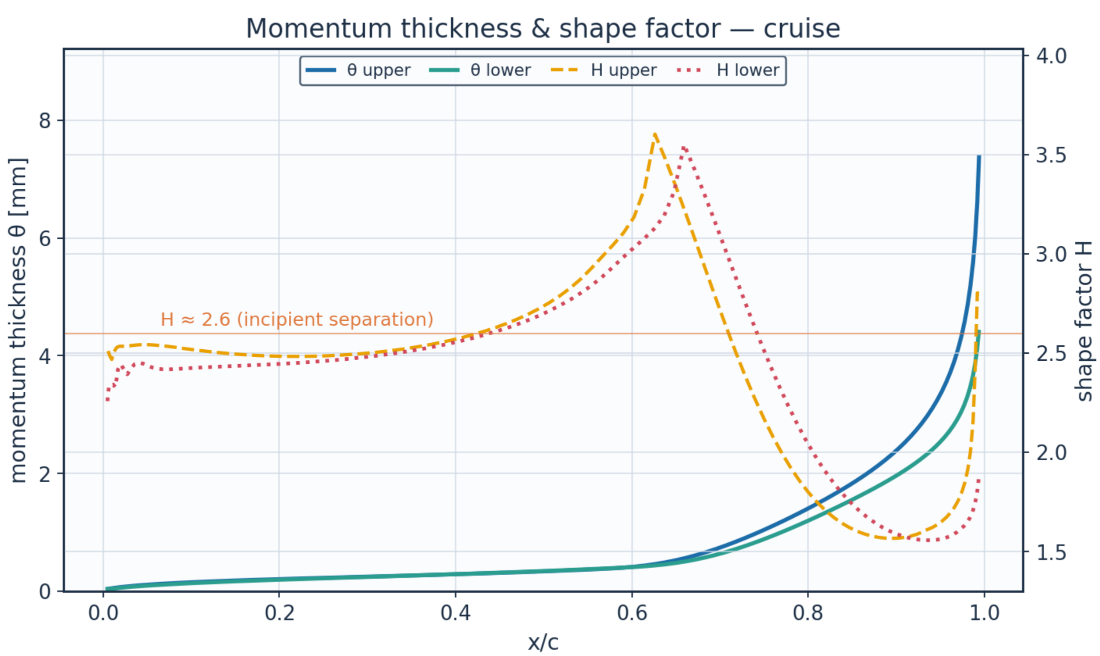


Fig. 15. Momentum thickness ϑ and shape factor H — cruise.

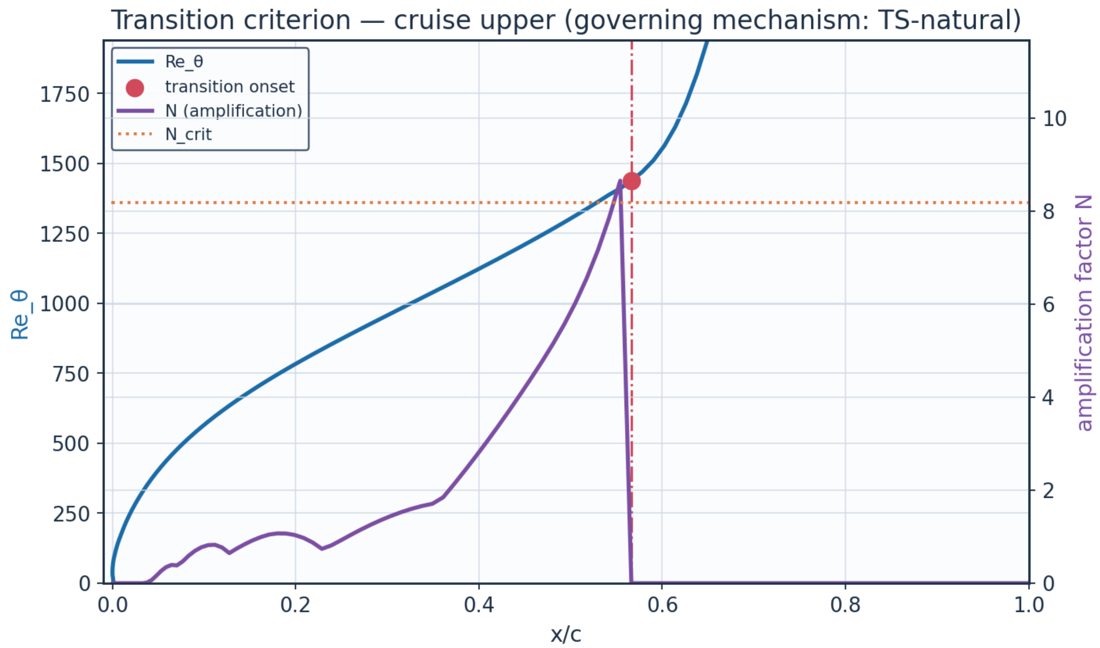


Fig. 16. The governing transition criterion — cruise. Two of the four branches are Reynolds-number thresholds and two are amplification integrals, so both pairs are drawn; at cruise it is N reaching N_crit that fires, and no Re_ϑt threshold is active at any station.

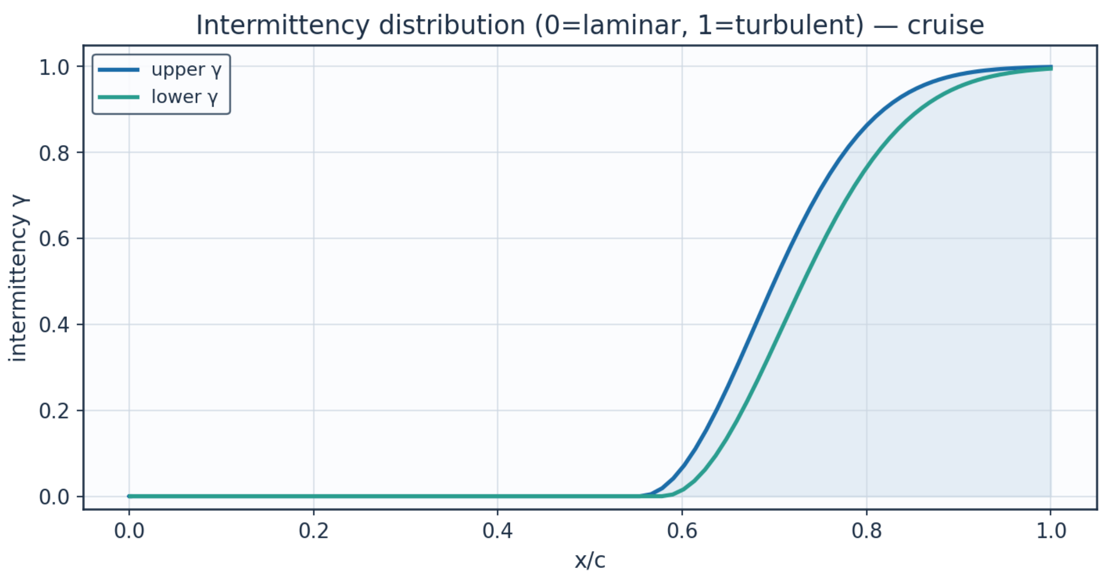


Fig. 17. Intermittency γ — cruise.

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm
0.001152	0.0	0.0	1.04845	0.6744	0.00544	1e-05
0.000243	0.014098	44712.0	0.53191	88.0777	0.71071	0.02631
0.003998	0.030357	96282.0	-0.19066	134.137	1.08237	0.03738
0.012383	0.052871	167687.0	-0.44319	146.3273	1.18073	0.05628
0.025317	0.083188	263841.0	-0.51616	149.6165	1.20727	0.07914
0.042683	0.121711	386021.0	-0.54817	151.0303	1.21868	0.10236

0.064323	0.168387	534059.0	-0.5751	152.2064	1.22817	0.12465
0.090042	0.222935	707065.0	-0.6054	153.5153	1.23873	0.14556
0.119612	0.284931	903693.0	-0.63998	154.9911	1.25064	0.16513
0.152769	0.353848	1122271.0	-0.67734	156.5653	1.26334	0.18358
0.189218	0.42908	1360877.0	-0.7151	158.1348	1.27601	0.20126
0.228631	0.509959	1617394.0	-0.75047	159.5867	1.28772	0.21855
0.27065	0.595771	1889557.0	-0.78059	160.8087	1.29758	0.23592
0.31489	0.685764	2174979.0	-0.80268	161.6976	1.30476	0.25384
0.360938	0.779156	2471183.0	-0.81436	162.1647	1.30852	0.27281
0.408355	0.87514	2775607.0	-0.81377	162.1411	1.30833	0.29338
0.456686	0.972886	3085621.0	-0.79978	161.581	1.30382	0.31617
0.505459	1.071544	3398527.0	-0.77203	160.4628	1.29479	0.34189
0.554191	1.170245	3711569.0	-0.73095	158.7879	1.28128	0.37147
0.6024	1.268103	4021937.0	-0.67765	156.5784	1.26345	0.41679
0.649607	1.364223	4326793.0	-0.61372	153.8722	1.24161	0.52579
0.695348	1.457706	4623285.0	-0.54105	150.7175	1.21616	0.7178
0.739176	1.547662	4908589.0	-0.46161	147.1669	1.18751	0.9722
0.780671	1.633218	5179941.0	-0.37727	143.2717	1.15608	1.25944
0.819445	1.713537	5434681.0	-0.28964	139.0764	1.12222	1.56255
0.855143	1.787825	5670295.0	-0.19999	134.6124	1.0862	1.88158
0.887448	1.855347	5884449.0	-0.10914	129.8925	1.04812	2.22899
0.916081	1.915435	6075027.0	-0.01737	124.9016	1.00784	2.62601
0.940801	1.967499	6240154.0	0.07566	119.5836	0.96493	3.10519
0.961407	2.01103	6378217.0	0.17118	113.8135	0.91837	3.72186

(columns 2-7 of 15; x_c repeated on each block)

x_c	**H_shape**	**Cf**	**Re_theta**	**Re_theta_ trans**	**n_factor**	**n_crit**
0.001152	2.61	2491.9080852	0.0		0.0	8.184
0.000243	2.1319	0.0138279	58.12		0.0	8.184
0.003998	2.4624	0.00428	122.45		0.0	8.184
0.012383	2.5058	0.0024818	199.37		0.0	8.184
0.025317	2.5343	0.0016657	285.95		0.0	8.184
0.042683	2.542	0.0012641	372.94		0.0666	8.184
0.064323	2.5379	0.0010366	457.27		0.3925	8.184
0.090042	2.5238	0.0008979	538.01		0.6962	8.184
0.119612	2.5076	0.000802	615.48		0.7682	8.184
0.152769	2.4938	0.0007282	690.34		0.9187	8.184
0.189218	2.4852	0.0006661	763.43		1.0684	8.184
0.228631	2.4834	0.00061	835.69		0.7385	8.184
0.27065	2.4897	0.0005566	908.11		1.1476	8.184
0.31489	2.506	0.0005039	981.75		1.5251	8.184
0.360938	2.5344	0.0004502	1057.77		1.8464	8.184
0.408355	2.5789	0.0003944	1137.39		3.035	8.184
0.456686	2.6469	0.0003347	1222.05		4.3971	8.184
0.505459	2.7511	0.000269	1313.51		6.0334	8.184
0.554191	2.9274	0.0001929	1410.36		8.6489	8.184
0.6024	3.1844	0.0003257	1563.3		0.0	8.184
0.649607	3.3423	0.0008082	1942.36		0.0	8.184
0.695348	2.7736	0.0014235	2603.93		0.0	8.184

0.739176	2.2735	0.0018989	3453.39		0.0	8.184
0.780671	1.9189	0.0021711	4368.44		0.0	8.184
0.819445	1.7078	0.0022652	5277.73		0.0	8.184
0.855143	1.6032	0.0022342	6171.43		0.0	8.184
0.887448	1.5676	0.0021217	7078.16		0.0	8.184
0.916081	1.5776	0.0019494	8045.86		0.0	8.184
0.940801	1.621	0.0017308	9140.79		0.0	8.184
0.961407	1.6948	0.0014754	10465.3		0.0	8.184

(columns 8-13 of 15; x_c repeated on each block)

x_c	**intermittency_gamma**	**state**
0.001152	0.0	laminar
0.000243	0.0	laminar
0.003998	0.0	laminar
0.012383	0.0	laminar
0.025317	0.0	laminar
0.042683	0.0	laminar
0.064323	0.0	laminar
0.090042	0.0	laminar
0.119612	0.0	laminar
0.152769	0.0	laminar
0.189218	0.0	laminar
0.228631	0.0	laminar
0.27065	0.0	laminar
0.31489	0.0	laminar
0.360938	0.0	laminar
0.408355	0.0	laminar
0.456686	0.0	laminar
0.505459	0.0	laminar
0.554191	0.0	transitional
0.6024	0.07223	transitional
0.649607	0.25516	transitional
0.695348	0.47635	transitional
0.739176	0.67214	transitional
0.780671	0.81327	transitional
0.819445	0.90082	transitional
0.855143	0.94951	transitional
0.887448	0.97464	transitional
0.916081	0.98706	transitional
0.940801	0.9931	turbulent
0.961407	0.99605	turbulent

Table 15. Cruise upper-surface BL state (surface_cruise_upper.csv, sampled). (columns 14-15 of 15; the table is split across 3 blocks so that no cell is compressed to illegibility, with x_c repeated on each)

(table sampled to 30 rows; full data in 04_solution/surface_cruise_upper.csv)

x_c	**arc_s_m**	**Re_x**	**Cp**	**Ue_ms**	**Ue_Uinf**	**theta_mm**
0.001152	0.0	0.0	1.04845	0.6744	0.00544	1e-05
0.006721	0.017559	55690.0	0.79679	61.87	0.49924	0.04063
0.016897	0.041927	132977.0	0.50913	89.9507	0.72582	0.05473
0.031587	0.074243	235470.0	0.32012	104.0654	0.83972	0.07483
0.05065	0.114743	363920.0	0.19326	112.4302	0.90721	0.0962
0.073903	0.163287	517883.0	0.10054	118.1126	0.95306	0.11713
0.101119	0.219531	696267.0	0.0275	122.3711	0.98743	0.13798

0.132034	0.282995	897550.0	-0.03295	125.7659	1.01482	0.15832
0.166346	0.353101	1119902.0	-0.08409	128.5536	1.03731	0.17816
0.203719	0.429198	1361253.0	-0.12717	130.846	1.05581	0.1977
0.243794	0.510579	1619360.0	-0.16236	132.6826	1.07063	0.21721
0.286186	0.596497	1891858.0	-0.1893	134.0676	1.08181	0.23703
0.330494	0.68618	2176300.0	-0.20746	134.9913	1.08926	0.25754
0.376307	0.778842	2470187.0	-0.21642	135.4443	1.09292	0.27913
0.423206	0.873685	2770995.0	-0.21605	135.4253	1.09276	0.30225
0.470769	0.969908	3076177.0	-0.20658	134.9465	1.0889	0.32732
0.518574	1.066701	3383168.0	-0.18866	134.035	1.08154	0.35482
0.566202	1.16325	3689382.0	-0.16332	132.7324	1.07103	0.38526
0.613236	1.258731	3992211.0	-0.13183	131.0908	1.05779	0.42303
0.659265	1.352314	4289022.0	-0.09556	129.1685	1.04228	0.50292
0.703882	1.443165	4577167.0	-0.05587	127.0246	1.02498	0.65369
0.746688	1.530451	4854003.0	-0.01398	124.7127	1.00632	0.86645
0.787291	1.613345	5116913.0	0.02917	122.276	0.98666	1.11461
0.825311	1.691043	5363340.0	0.07296	119.7418	0.96621	1.37424
0.860385	1.762769	5590828.0	0.11718	117.1168	0.94503	1.63365
0.892165	1.827793	5797058.0	0.16204	114.3808	0.92295	1.89308
0.920332	1.885439	5979891.0	0.20828	111.4777	0.89953	2.16171
0.944596	1.935102	6137401.0	0.25731	108.2997	0.87388	2.45722
0.964701	1.976252	6267912.0	0.31158	104.6523	0.84445	2.81192
0.980432	2.008448	6370025.0	0.37537	100.1704	0.80829	3.29473

(columns 2-7 of 15; x_c repeated on each block)

x_c	**H_shape**	**Cf**	**Re_theta**	**Re_theta_ trans**	**n_factor**	**n_crit**
0.001152	2.61	2491.9079902	0.0		0.0	8.184
0.006721	2.3373	0.0096899	63.69		0.0	8.184
0.016897	2.4177	0.0045057	123.35		0.0	8.184
0.031587	2.4227	0.0028505	193.72		0.0	8.184
0.05065	2.4362	0.0020257	267.8		0.0	8.184
0.073903	2.4177	0.001628	341.41		0.0	8.184
0.101119	2.4245	0.0013256	415.58		0.0	8.184
0.132034	2.4315	0.0011163	488.98		0.0	8.184
0.166346	2.438	0.0009639	561.44		0.0	8.184
0.203719	2.446	0.0008458	633.16		0.0	8.184
0.243794	2.4575	0.0007487	704.54		0.0	8.184
0.286186	2.4743	0.0006647	776.11		0.0842	8.184
0.330494	2.4985	0.0005888	848.53		0.5034	8.184
0.376307	2.5325	0.0005176	922.48		1.1683	8.184
0.423206	2.5797	0.0004486	998.74		2.0429	8.184
0.470769	2.6461	0.0003797	1078.12		3.5186	8.184
0.518574	2.7401	0.000309	1161.53		5.2781	8.184
0.566202	2.883	0.0002334	1250.04		7.6134	8.184
0.613236	3.0736	0.0002625	1354.6		0.0	8.184
0.659265	3.5491	0.0005794	1588.92		0.0	8.184
0.703882	3.0469	0.0011622	2033.97		0.0	8.184
0.746688	2.548	0.001688	2651.06		0.0	8.184

0.787291	2.1457	0.0020627	3349.1		0.0	8.184
0.825311	1.8671	0.0022753	4050.32		0.0	8.184
0.860385	1.6965	0.0023576	4717.2		0.0	8.184
0.892165	1.604	0.0023491	5347.74		0.0	8.184
0.920332	1.5635	0.002278	5962.12		0.0	8.184
0.944596	1.5587	0.0021551	6596.34		0.0	8.184
0.964701	1.5857	0.001971	7309.62		0.0	8.184
0.980432	1.6476	0.0017154	8218.27		0.0	8.184

(columns 8-13 of 15; x_c repeated on each block)

x_c	intermittency_gamma	state
0.001152	0.0	laminar
0.006721	0.0	laminar
0.016897	0.0	laminar
0.031587	0.0	laminar
0.05065	0.0	laminar
0.073903	0.0	laminar
0.101119	0.0	laminar
0.132034	0.0	laminar
0.166346	0.0	laminar
0.203719	0.0	laminar
0.243794	0.0	laminar
0.286186	0.0	laminar
0.330494	0.0	laminar
0.376307	0.0	laminar
0.423206	0.0	laminar
0.470769	0.0	laminar
0.518574	0.0	laminar
0.566202	0.0	laminar
0.613236	0.03546	transitional
0.659265	0.17513	transitional
0.703882	0.37048	transitional
0.746688	0.56496	transitional
0.787291	0.72278	transitional
0.825311	0.83362	transitional
0.860385	0.90372	transitional
0.892165	0.94494	transitional
0.920332	0.96808	transitional
0.944596	0.98079	transitional
0.964701	0.98771	transitional
0.980432	0.99148	turbulent

Table 16. Cruise lower-surface BL state (surface_cruise_lower.csv, sampled). (columns 14-15 of 15; the table is split across 3 blocks so that no cell is compressed to illegibility, with x_c repeated on each)

(table sampled to 30 rows; full data in 04_solution/surface_cruise_lower.csv)

9.4 Surface distributions — climb (off-design, elevated Tu)

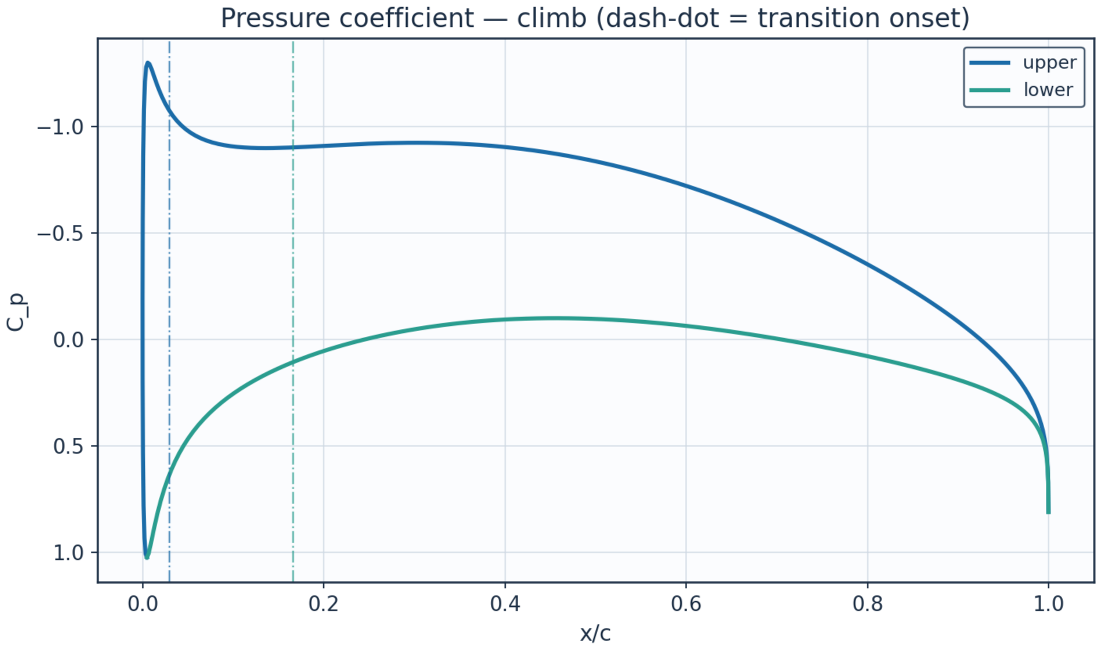


Fig. 18. Pressure coefficient C_p — climb.

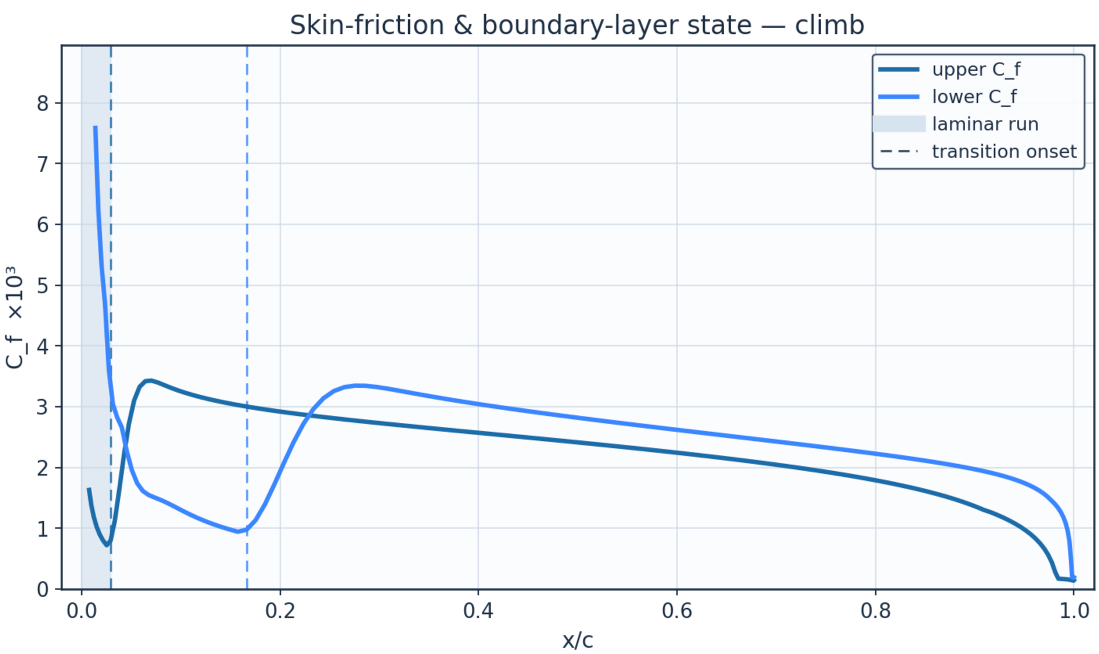


Fig. 19. Skin-friction C_f and laminar run — climb.

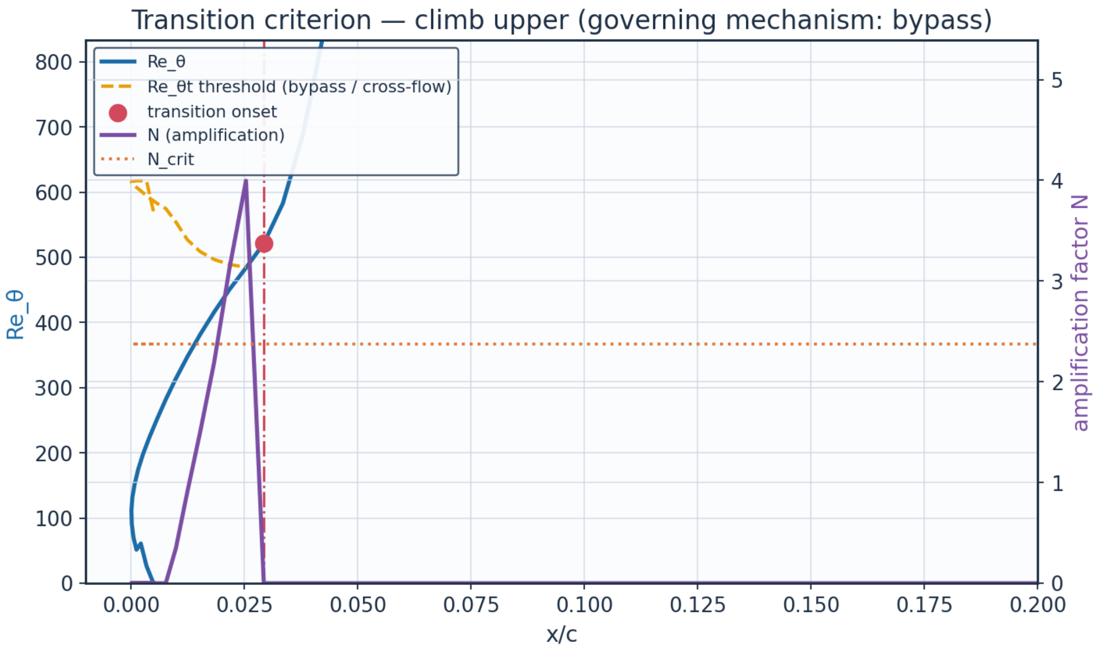


Fig. 19b. The governing transition criterion — climb: here Re_ϑ crosses the falling Abu-Ghannam & Shaw bypass threshold while N is still far below N_crit.

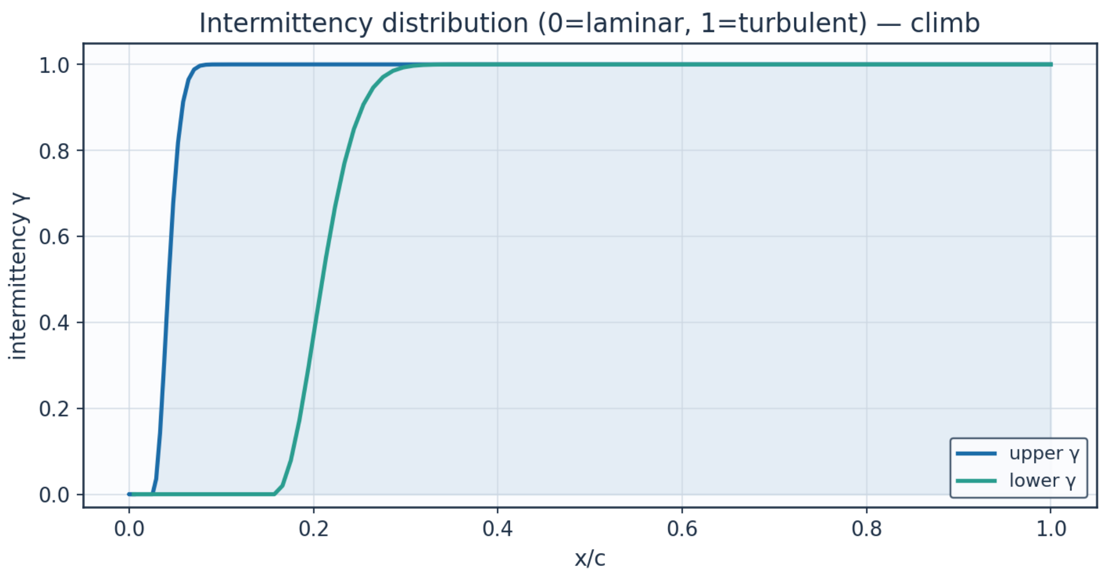


Fig. 20. Intermittency γ — climb.

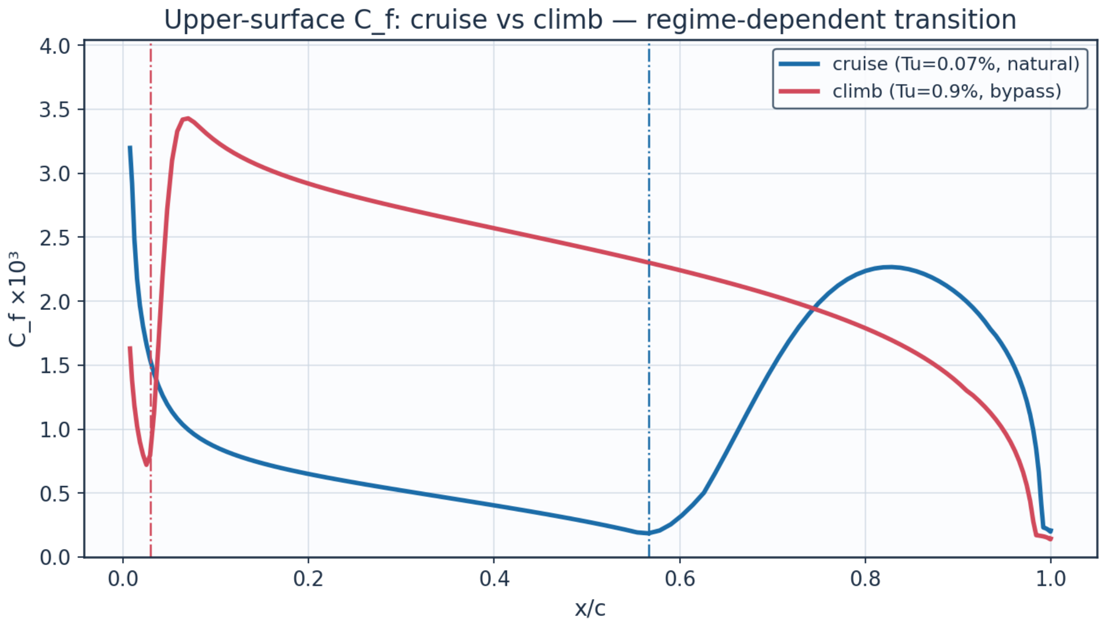


Fig. 21. C_f cruise vs climb — regime-dependent transition.

9.5 Aerodynamic polars

alpha_deg	Cl	Cd	L_over_D	xtr_upper_c	xtr_lower_c
-3.0	-0.1114	0.00563	-19.8	0.673	0.149
-2.0	0.0287	0.00453	6.3	0.65	0.4
-1.0	0.1683	0.00434	38.8	0.626	0.471
0.0	0.3077	0.00424	72.6	0.602	0.531
1.0	0.4472	0.00433	103.3	0.566	0.566
2.0	0.5872	0.00446	131.8	0.53	0.602
3.0	0.7281	0.0052	140.0	0.408	0.625
4.0	0.8704	0.00762	114.2	0.048	0.659
5.0	1.0146	0.00828	122.6	0.025	0.682
6.0	1.1612	0.00888	130.7	0.018	0.715
7.0	1.3111	0.00951	137.8	0.022	0.736
8.0	1.4654	0.01042	140.6	0.015	0.767

Table 17. Aerodynamic polar (aero_polar.csv).

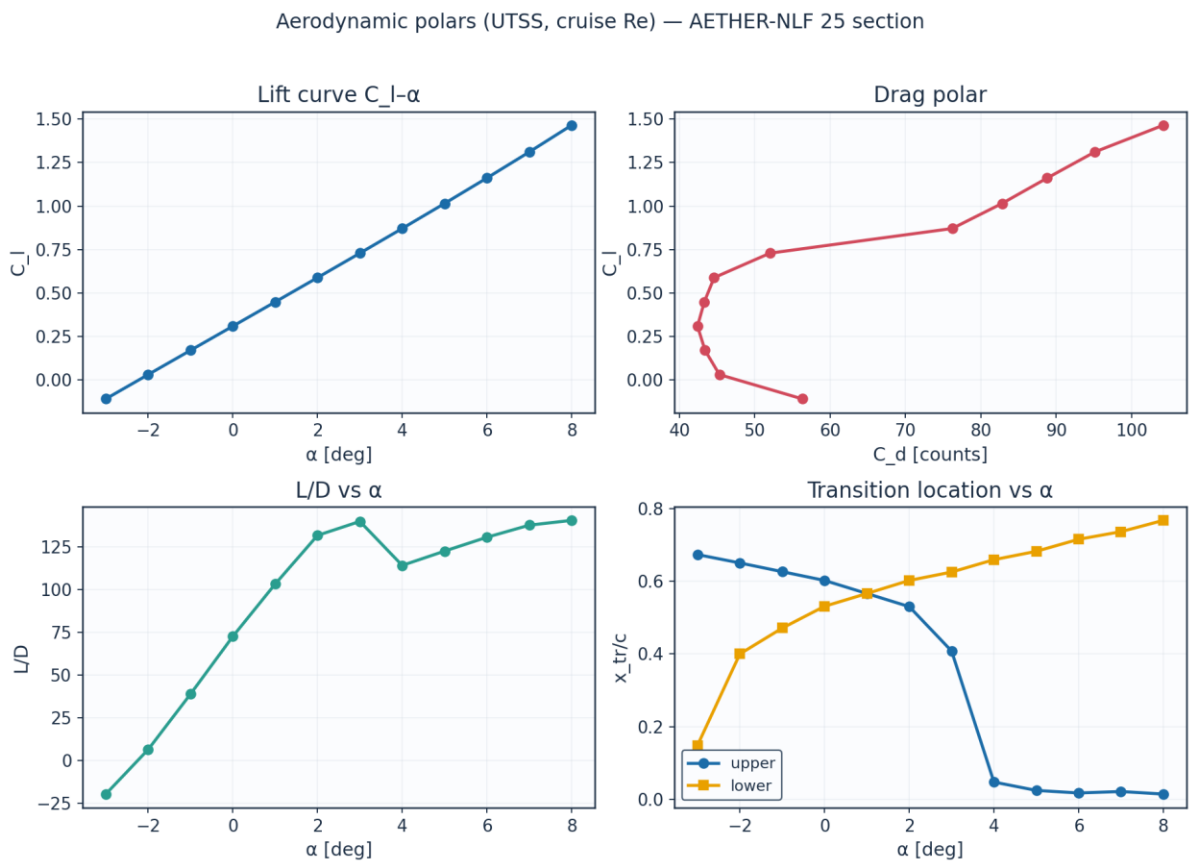


Fig. 22. Lift curve, drag polar, L/D and transition vs angle of attack.

9.6 Span-wise distribution (3-D)

eta	y_m	chord_m	Re_local	alpha_eff_deg	xtr_upper_c	xtr_lower_c
0.0	0.0	2.6	8246200.0	1.5	0.53	0.566
0.089	0.757	2.484	7878900.0	1.23	0.542	0.554
0.178	1.515	2.368	7511500.0	0.97	0.554	0.554
0.267	2.272	2.253	7144200.0	0.7	0.566	0.542
0.356	3.029	2.137	6776900.0	0.43	0.578	0.542
0.445	3.786	2.021	6409500.0	0.16	0.59	0.531
0.535	4.544	1.905	6042200.0	-0.1	0.602	0.531
0.624	5.301	1.789	5674900.0	-0.37	0.614	0.519
0.713	6.058	1.673	5307600.0	-0.64	0.626	0.519
0.802	6.815	1.558	4940200.0	-0.91	0.684	0.507
0.891	7.573	1.442	4572900.0	-1.17	0.695	0.495
0.98	8.33	1.326	4205600.0	-1.44	0.707	0.495

(columns 2-7 of 9; eta repeated on each block)

eta	Cd_section	laminar_fraction
0.0	0.0043	0.548
0.089	0.00431	0.548
0.178	0.00428	0.554
0.267	0.0043	0.554
0.356	0.00428	0.56

0.445	0.00431	0.56
0.535	0.0043	0.566
0.624	0.00434	0.566
0.713	0.00434	0.572
0.802	0.00417	0.595
0.891	0.00426	0.595
0.98	0.00432	0.601

Table 18. Span-wise distribution (spanwise_distribution.csv). (columns 8-9 of 9; the table is split across 2 blocks so that no cell is compressed to illegibility, with eta repeated on each)

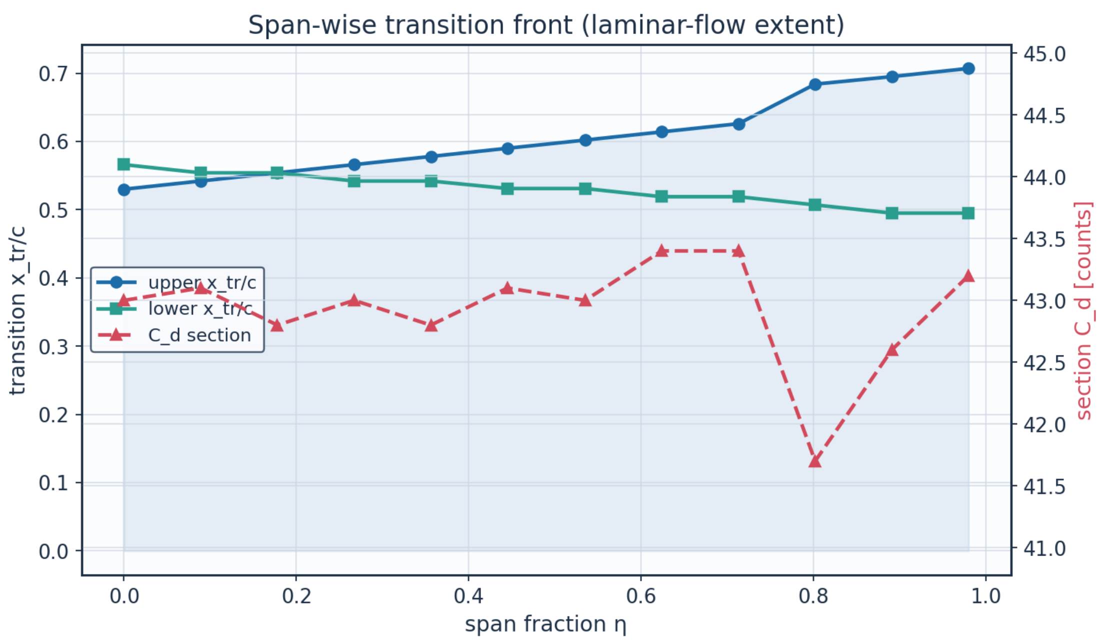


Fig. 23. Span-wise transition front and section drag.

10. Post-Processing — Contours, Profiles and 3-D Fields

10.1 Pressure and velocity contours

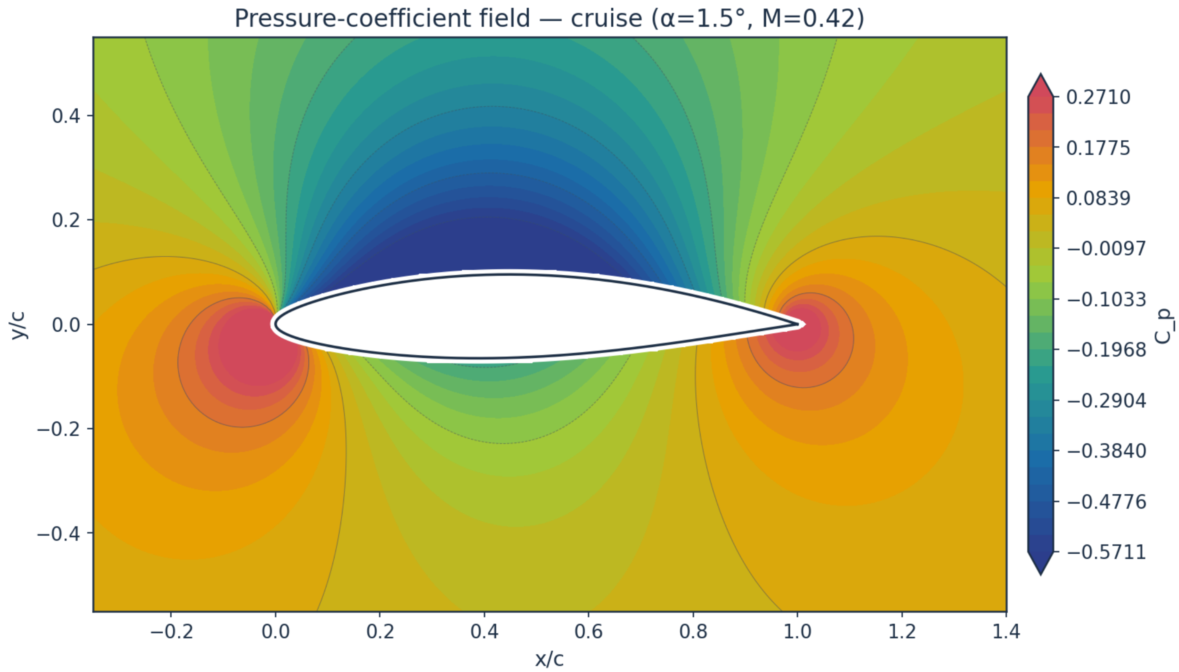


Fig. 24. Pressure-coefficient contour — cruise.

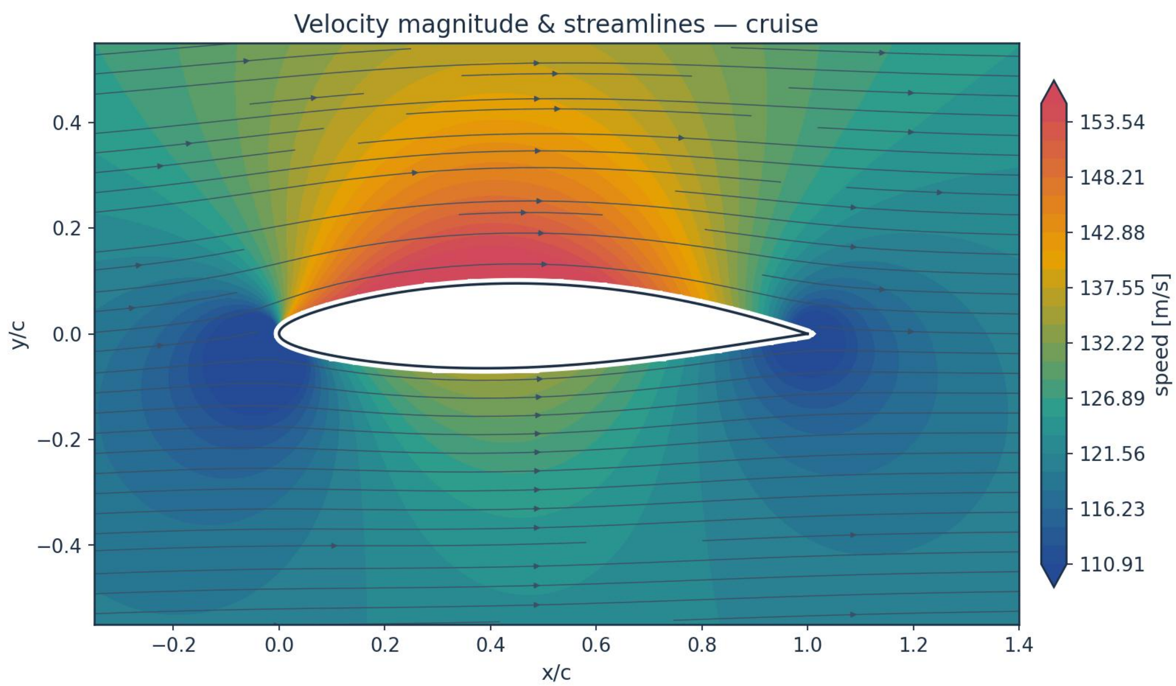


Fig. 25. Velocity magnitude and streamlines — cruise.

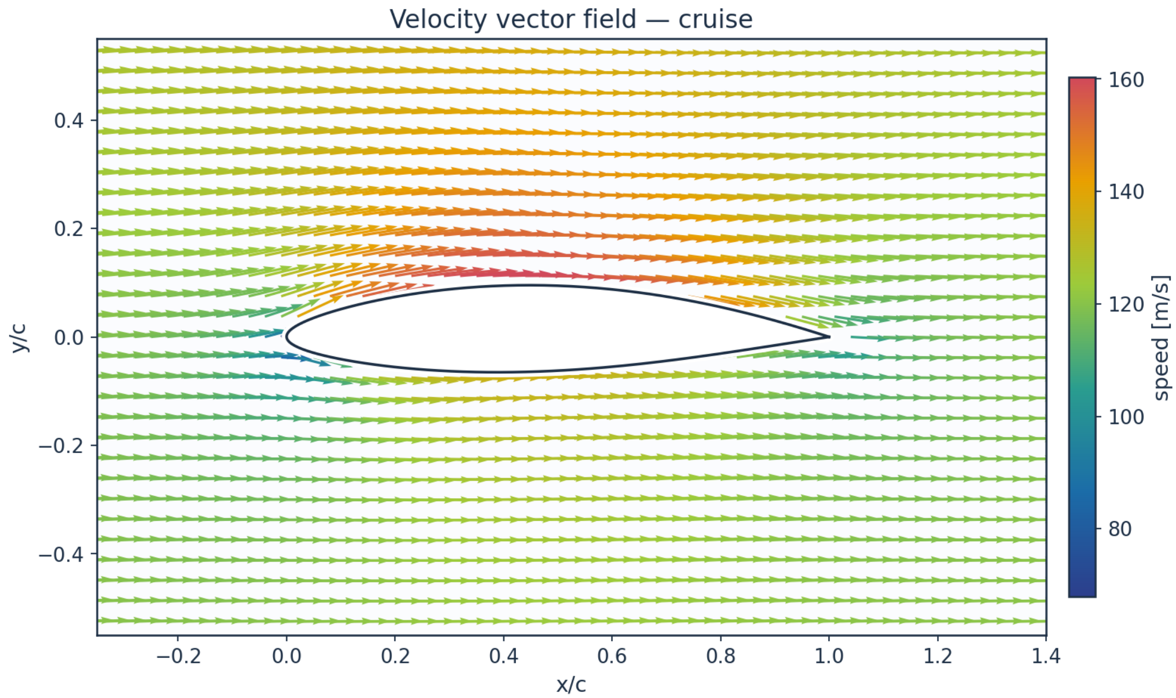


Fig. 26. Velocity vector field — cruise.

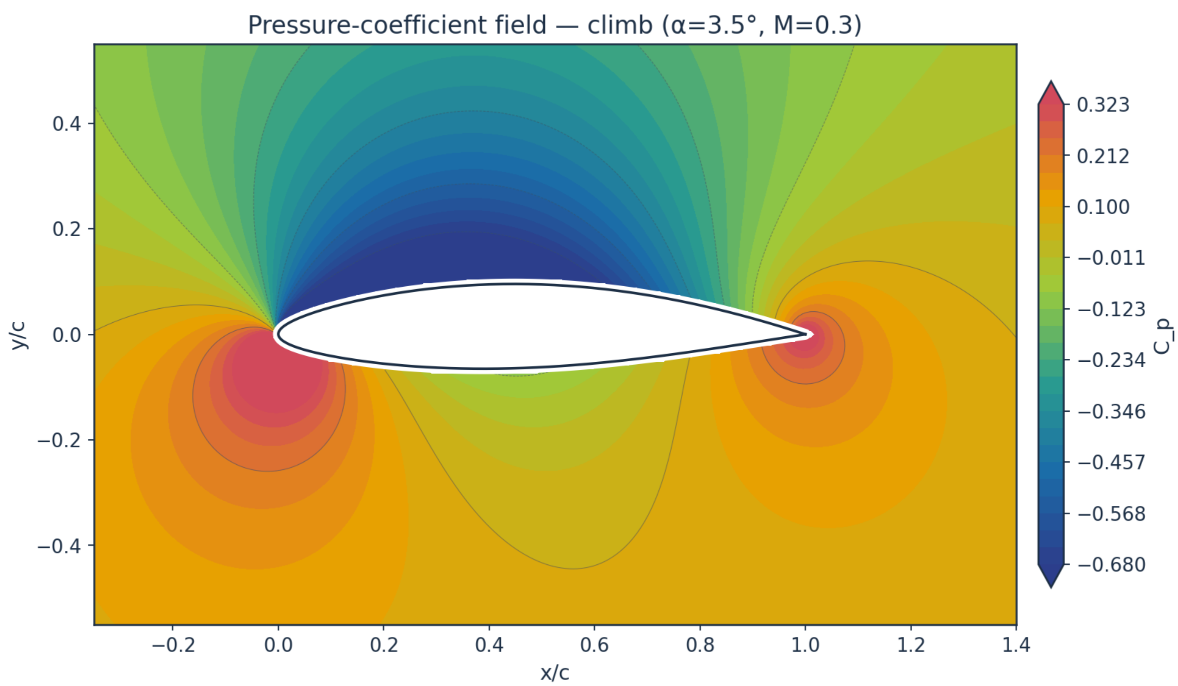


Fig. 27. Pressure-coefficient contour — climb.

10.2 Boundary-layer velocity and temperature profiles

The profiles are reconstructed from the marched state and not from an assumed shape. The laminar leg is the Falkner-Skan profile at the shape factor the march solved for, read from the same family that supplies the closure functions; the turbulent leg is the power law that shape factor implies, $H = (n+2)/n$; and the two are blended by the same intermittency that blends C_f, θ and H, each on its own thickness — $\delta_{99} = \eta_{99}\, \theta/\theta_\eta$ for the similarity profile and $\delta = \theta(n+1)(n+2)/n$ for the power law. Temperature profiles then follow from the compressible Crocco–Busemann relation (Eq. E21), showing wall-recovery heating through the boundary layer.

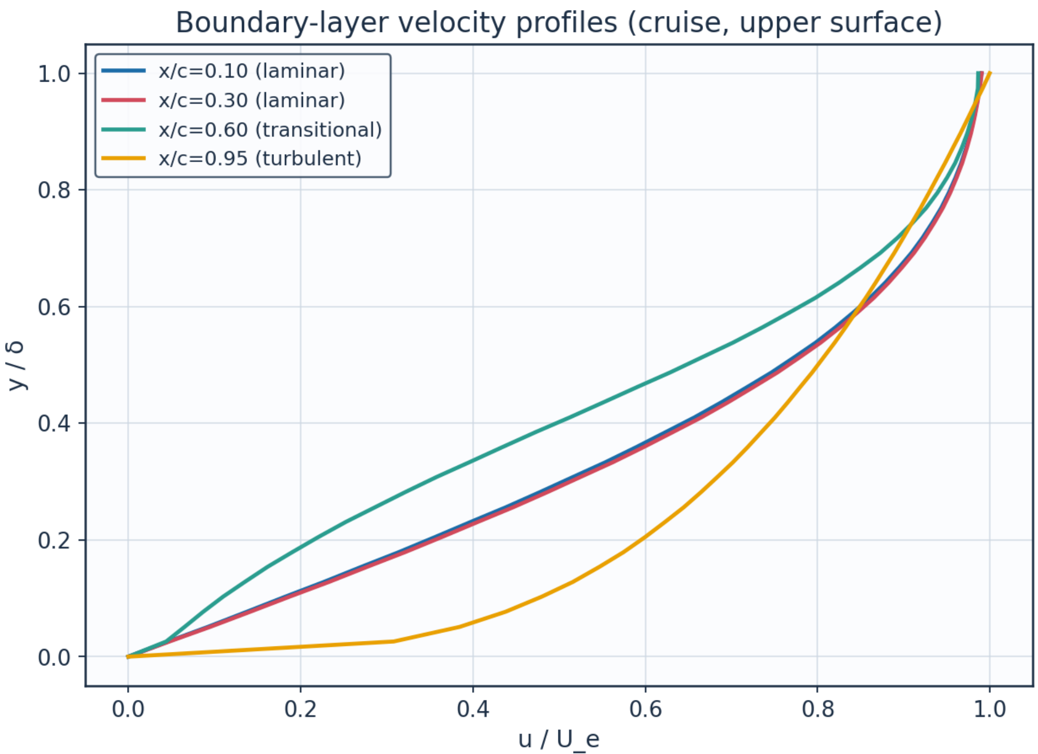


Fig. 28. Boundary-layer velocity profiles.

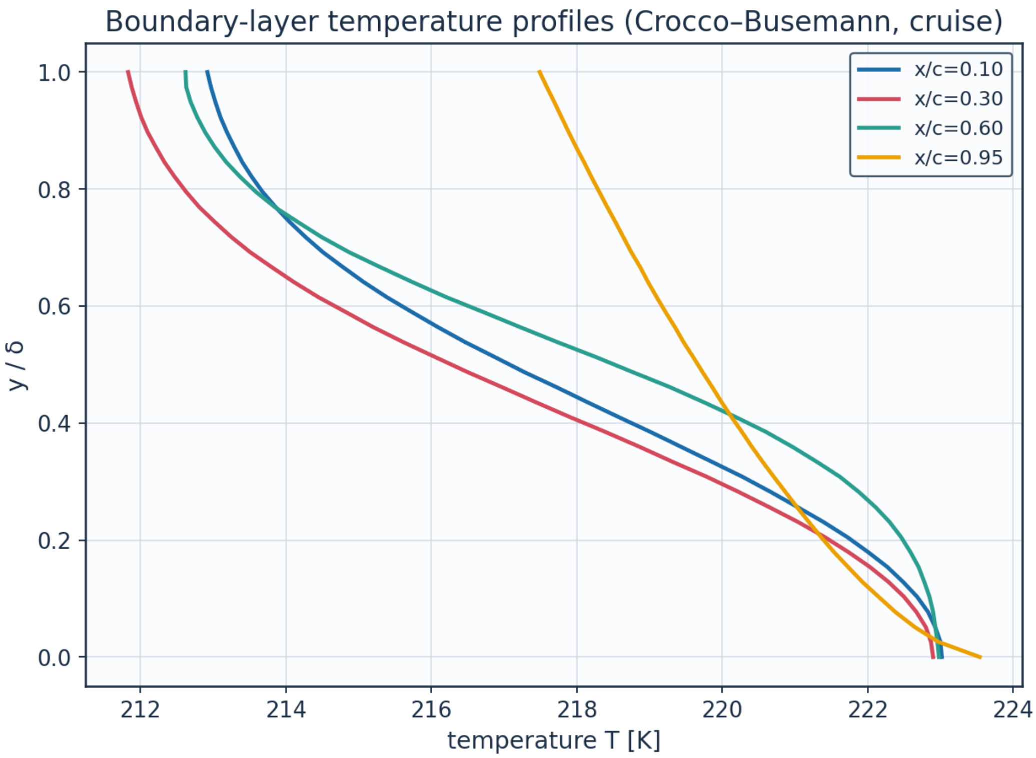


Fig. 29. Boundary-layer temperature profiles (K).

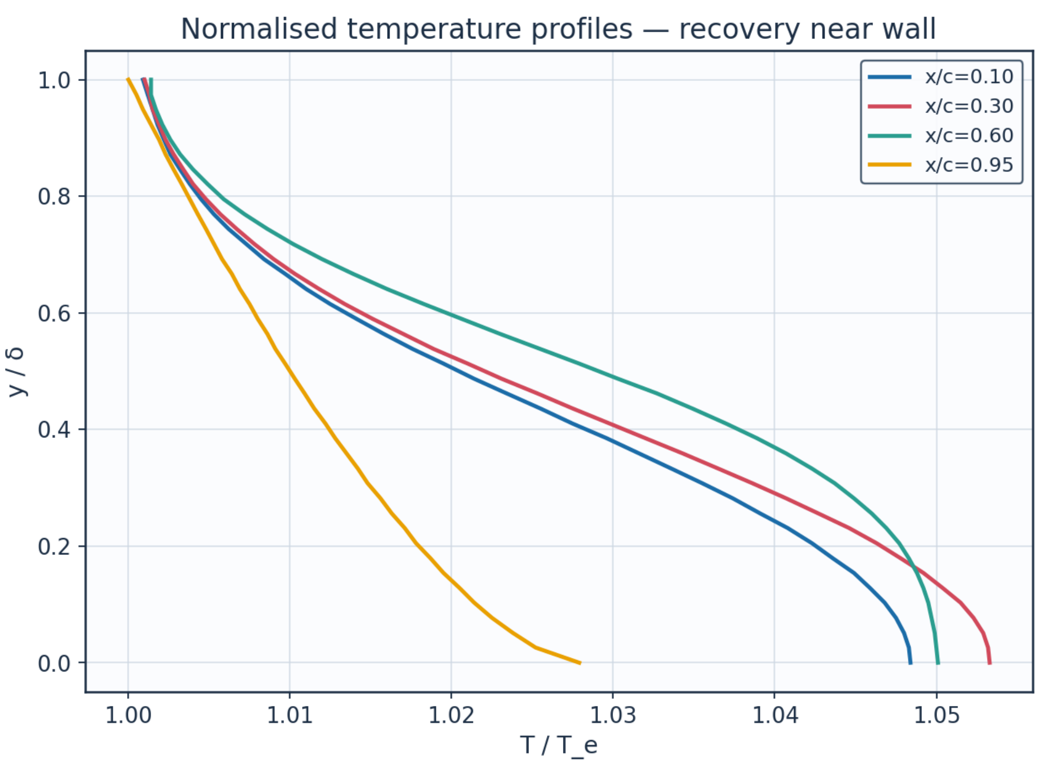


Fig. 30. Normalised temperature profiles T/T_e.

station	x_c	y_delta	y_mm	u_Ue	T_Te	T_K
x/c=0.10	0.1	0.0	0.0	0.0	1.0484	223.02
x/c=0.10	0.1	0.128	0.1442	0.2268	1.0459	222.49
x/c=0.10	0.1	0.256	0.2885	0.4381	1.0391	221.04
x/c=0.10	0.1	0.385	0.4327	0.6239	1.0296	219.01
x/c=0.10	0.1	0.513	0.5769	0.7728	1.0195	216.87
x/c=0.10	0.1	0.641	0.7211	0.8785	1.011	215.07
x/c=0.10	0.1	0.769	0.8654	0.9434	1.0053	213.86
x/c=0.10	0.1	0.897	1.0096	0.9773	1.0022	213.19
x/c=0.30	0.3	0.0	0.0	0.0	1.0533	222.9
x/c=0.30	0.3	0.128	0.2403	0.2327	1.0504	222.29
x/c=0.30	0.3	0.256	0.4806	0.4465	1.0427	220.65
x/c=0.30	0.3	0.385	0.721	0.6321	1.032	218.39
x/c=0.30	0.3	0.513	0.9613	0.7791	1.021	216.05
x/c=0.30	0.3	0.641	1.2016	0.8824	1.0118	214.11
x/c=0.30	0.3	0.769	1.4419	0.9453	1.0057	212.81
x/c=0.30	0.3	0.897	1.6822	0.978	1.0023	212.1
x/c=0.60	0.6	0.0	0.0	0.0	1.0501	222.98
x/c=0.60	0.6	0.128	0.4079	0.135	1.0492	222.78
x/c=0.60	0.6	0.256	0.8158	0.2862	1.046	222.11
x/c=0.60	0.6	0.385	1.2236	0.4729	1.0389	220.6
x/c=0.60	0.6	0.513	1.6315	0.6657	1.0279	218.26
x/c=0.60	0.6	0.641	2.0394	0.8249	1.016	215.74
x/c=0.60	0.6	0.769	2.4473	0.9259	1.0072	213.85
x/c=0.60	0.6	0.897	2.8552	0.9737	1.0026	212.89
x/c=0.95	0.95	0.0	0.0	0.0	1.0279	223.54
x/c=0.95	0.95	0.128	2.9437	0.5158	1.0205	221.93
x/c=0.95	0.95	0.256	5.8874	0.6446	1.0163	221.03
x/c=0.95	0.95	0.385	8.831	0.7349	1.0128	220.27

(columns 2-7 of 12; station repeated on each block)

station	Me_edge	H_shape	delta_mm	intermittency_gamma	state
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.30	0.547	2.501	1.8745	0.0	laminar
x/c=0.30	0.547	2.501	1.8745	0.0	laminar
x/c=0.30	0.547	2.501	1.8745	0.0	laminar
x/c=0.30	0.547	2.501	1.8745	0.0	laminar
x/c=0.30	0.547	2.501	1.8745	0.0	laminar
x/c=0.30	0.547	2.501	1.8745	0.0	laminar
x/c=0.30	0.547	2.501	1.8745	0.0	laminar
x/c=0.30	0.547	2.501	1.8745	0.0	laminar
x/c=0.60	0.531	3.184	3.1815	0.072	transitional
x/c=0.60	0.531	3.184	3.1815	0.072	transitional
x/c=0.60	0.531	3.184	3.1815	0.072	transitional
x/c=0.60	0.531	3.184	3.1815	0.072	transitional
x/c=0.60	0.531	3.184	3.1815	0.072	transitional
x/c=0.60	0.531	3.184	3.1815	0.072	transitional
x/c=0.60	0.531	3.184	3.1815	0.072	transitional
x/c=0.60	0.531	3.184	3.1815	0.072	transitional

x/c=0.95	0.396	1.652	22.9607	0.995	turbulent
x/c=0.95	0.396	1.652	22.9607	0.995	turbulent
x/c=0.95	0.396	1.652	22.9607	0.995	turbulent
x/c=0.95	0.396	1.652	22.9607	0.995	turbulent

Table 19. BL velocity/temperature profiles (bl_profiles_cruise.csv, sampled). (columns 8-12 of 12; the table is split across 2 blocks so that no cell is compressed to illegibility, with station repeated on each)

(table sampled to 28 rows; full data in 04_solution/bl_profiles_cruise.csv)

10.3 Three-dimensional surface contours and vectors

The full 3-D wing surface is coloured by the predicted fields; the transition front is directly visible as the laminar-to-turbulent boundary on the intermittency contour.

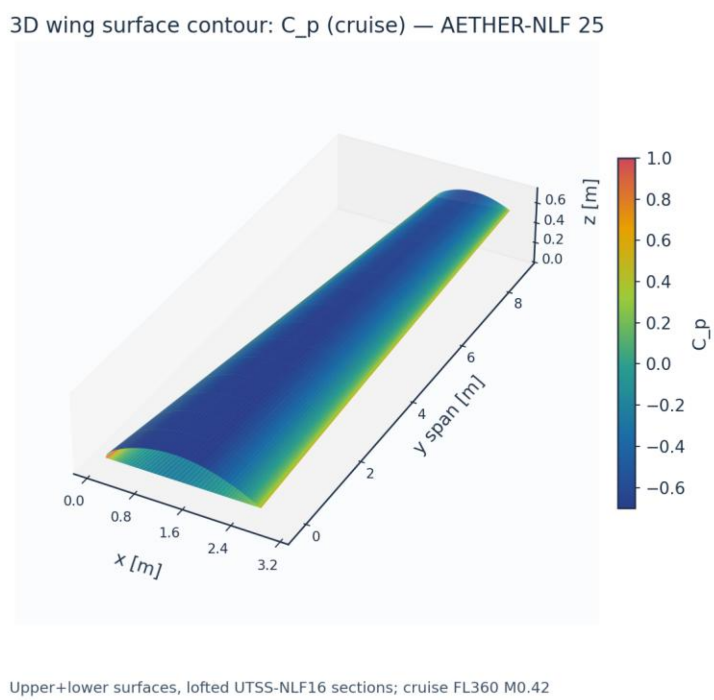


Fig. 31. 3-D wing surface contour — pressure C_p.

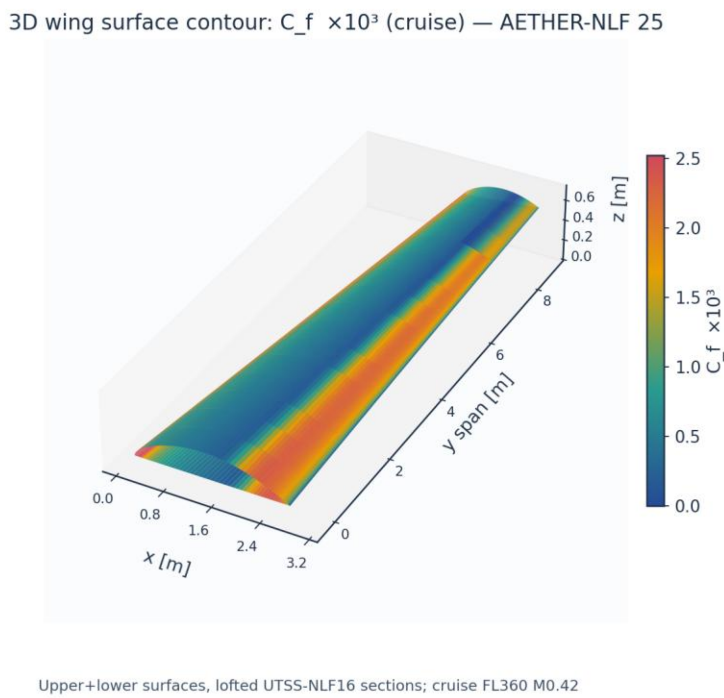


Fig. 32. 3-D wing surface contour — skin friction C_f (×10³).

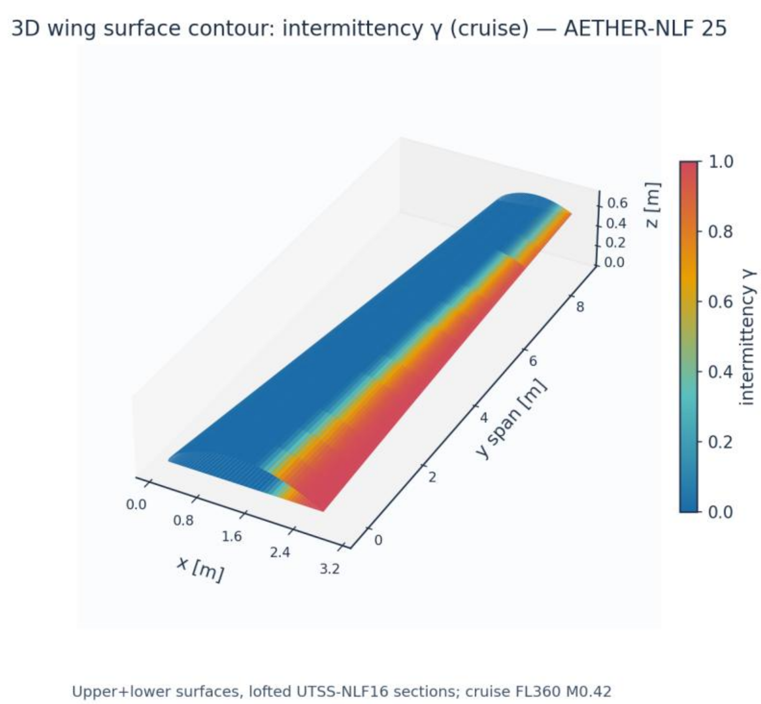


Fig. 33. 3-D wing surface contour — intermittency γ (transition front).

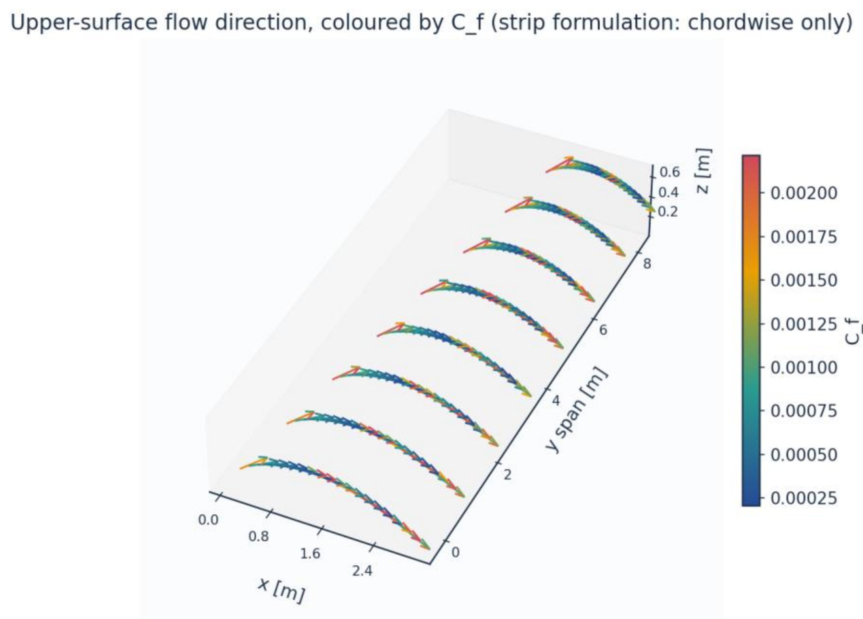


Fig. 34. Upper-surface flow direction coloured by C_f (chordwise only — the strip formulation carries no span-wise wall shear).

11. Validation and Calibration

To qualify as universal, the solver is validated against every published dataset the kernel's four branches reach — five flat plates spanning 0.03 to 6 per cent free-stream turbulence (bypass, natural and separation-induced transition), 86 aerofoil conditions on the NLF(1)-0416 section, and two swept wings from different facilities and eras — using ONE universal calibration set with no per-case re-tuning of the physics. The transition-onset momentum-thickness Reynolds number Re_θt is the metric on the plates and the transition location x_tr/c on the aerofoil and the swept wings. The skin-friction error is reported separately over the laminar run and over the turbulent run, on the stations where the measurement and the prediction are in the same state; pooled across transition it measures the onset error a second time, in the wrong units, because a plate whose onset is early by 16 % is then charged with the whole laminar-to-turbulent step in C_f over the interval between the two onsets.

case	Tu_inlet_pct	L_turb_mm	Re_theta_t_exp	Re_theta_t_pred	Re_theta_t_err_pct	Re_x_tr_exp
ERCOFTAC T3A flat plate (bypass transition)	3.043	1.5299999999999998	272.3	282.1	3.6	1.348e+05
ERCOFTAC T3A- flat plate (low-Tu	0.874	0.98	818.8	683.6	-16.5	1.443e+06

bypass)						
ERCOFTAC T3B flat plate (high-Tu bypass)	5.952	3.83	181.3	168.0	-7.3	5.910e+04
ERCOFTAC T3C4 flat plate (laminar separation bubble)	2.11		381.3	271.4	-28.8	1.785e+05
Schubauer & Skramstad flat plate (natural transition)	0.03		1100.0	1161.6	5.6	2.800e+06

(columns 2-7 of 15; case repeated on each block)

case	Re_x_tr_pred	Cf_err_ laminar_pct	n_pts_laminar	Cf_err_ turbulent_pct	n_pts_ turbulent	Cf_err_all_pts_ pct
ERCOFTAC T3A flat plate (bypass transition)	1.804e+05	3.8	4	4.8	3	17.2
ERCOFTAC T3A- flat plate (low-Tu bypass)	1.060e+06	4.2	8		0	121.6
ERCOFTAC T3B flat plate (high-Tu bypass)	6.405e+04	8.2	3	8.2	9	12.4
ERCOFTAC T3C4 flat plate (laminar separation bubble)	1.692e+05	21.9	9		0	37.9
Schubauer & Skramstad flat plate (natural transition)	3.060e+06		0		0	

(columns 8-13 of 15; case repeated on each block)

case	Re_theta_meas_over_blasius	mechanism
ERCOFTAC T3A flat plate (bypass transition)	1.117	bypass
ERCOFTAC T3A- flat plate (low-Tu bypass)	1.027	bypass
ERCOFTAC T3B flat plate (high-Tu bypass)	1.123	bypass
ERCOFTAC T3C4 flat plate (laminar separation bubble)	1.359	separation
Schubauer & Skramstad flat plate (natural transition)		TS-natural

Table 20. Validation summary — predicted vs published Re_ϑt. (columns 14-15 of 15; the table is split across 3 blocks so that no cell is compressed to illegibility, with case repeated on each)

11.1 Flat plates

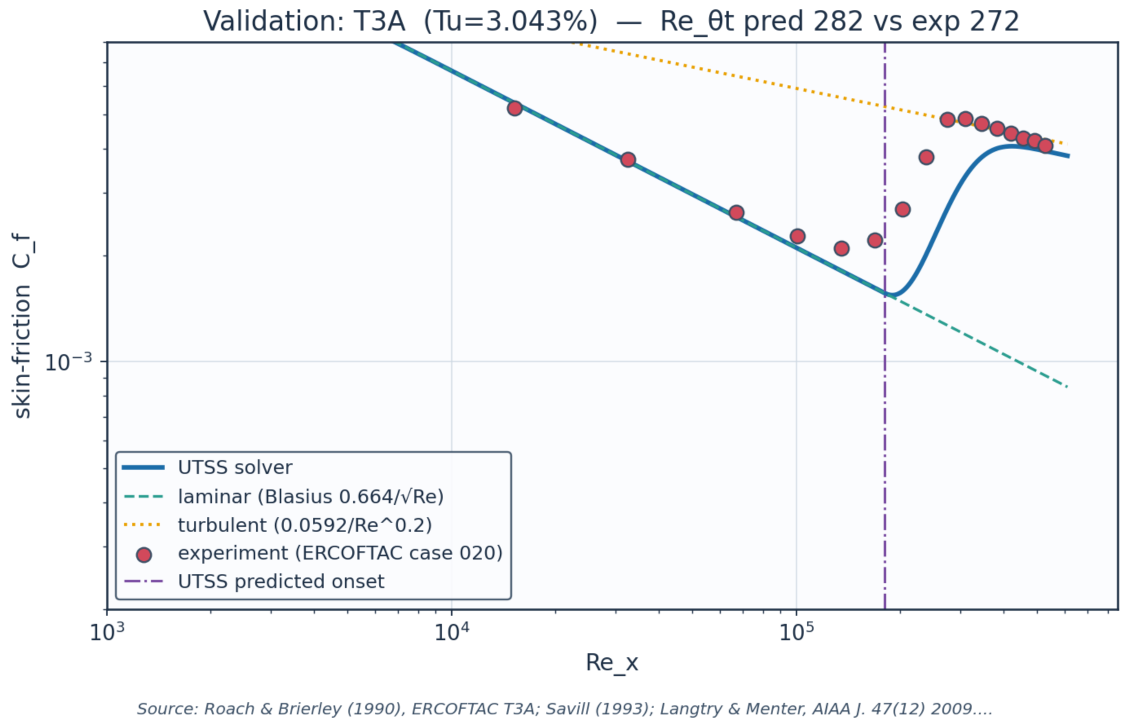


Fig. 35. Validation — ERCOFTAC T3A flat plate (Tu = 3.0 %, bypass).

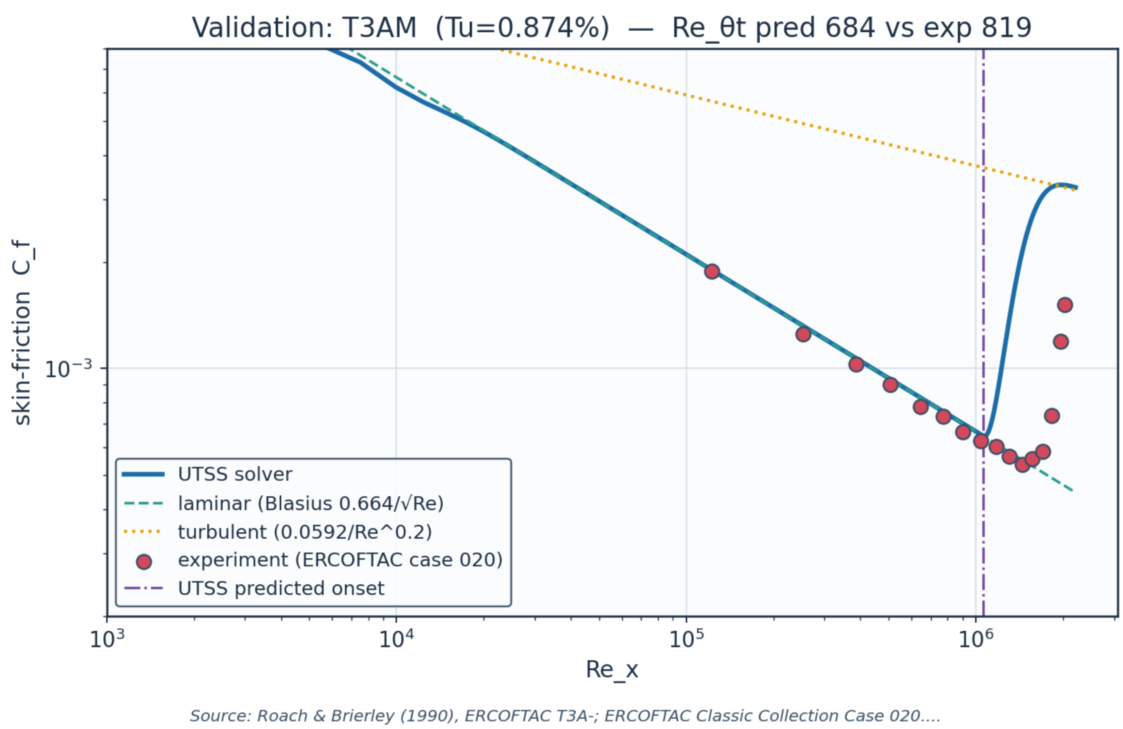
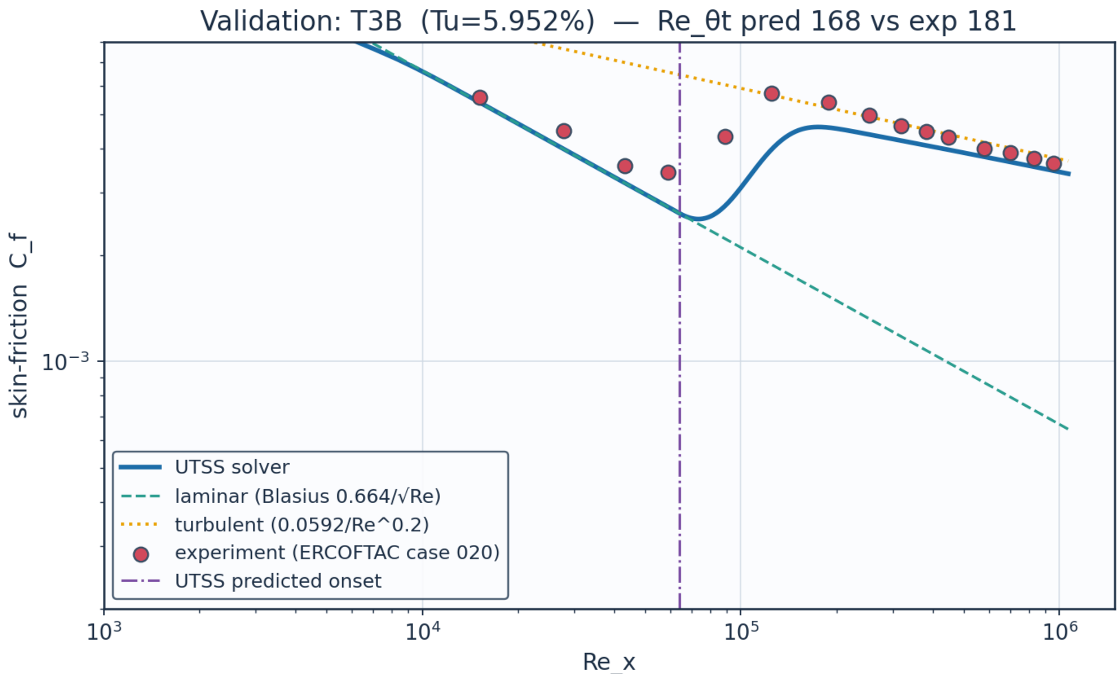
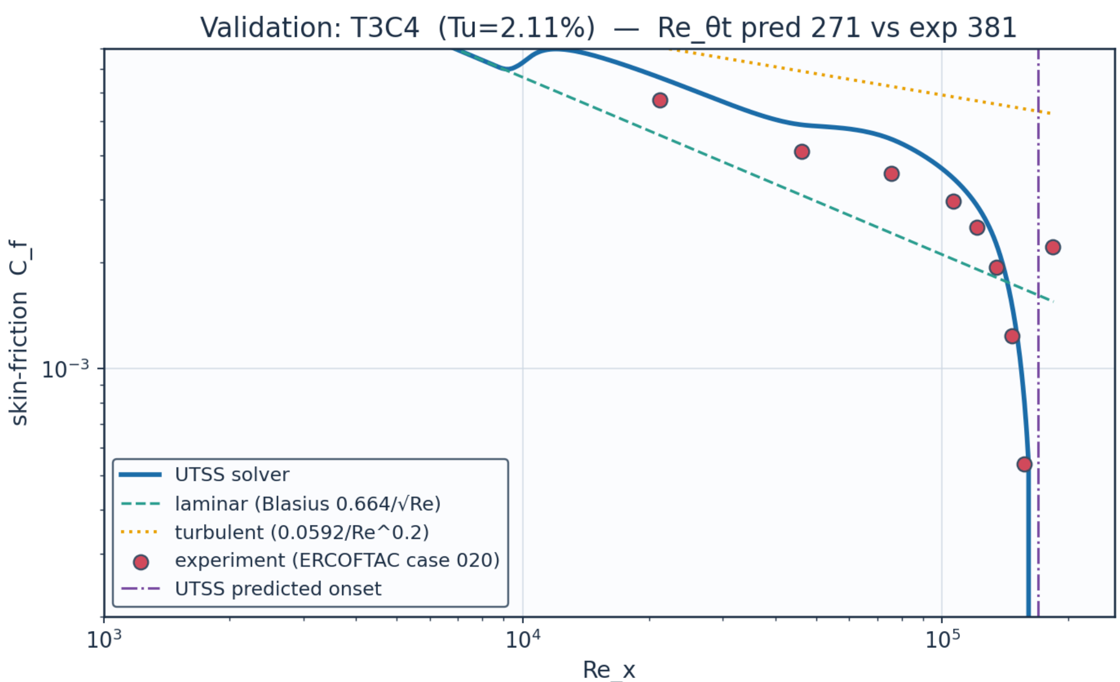


Fig. 36. Validation — ERCOFTAC T3A⁻ flat plate (Tu = 0.87 %, bypass).



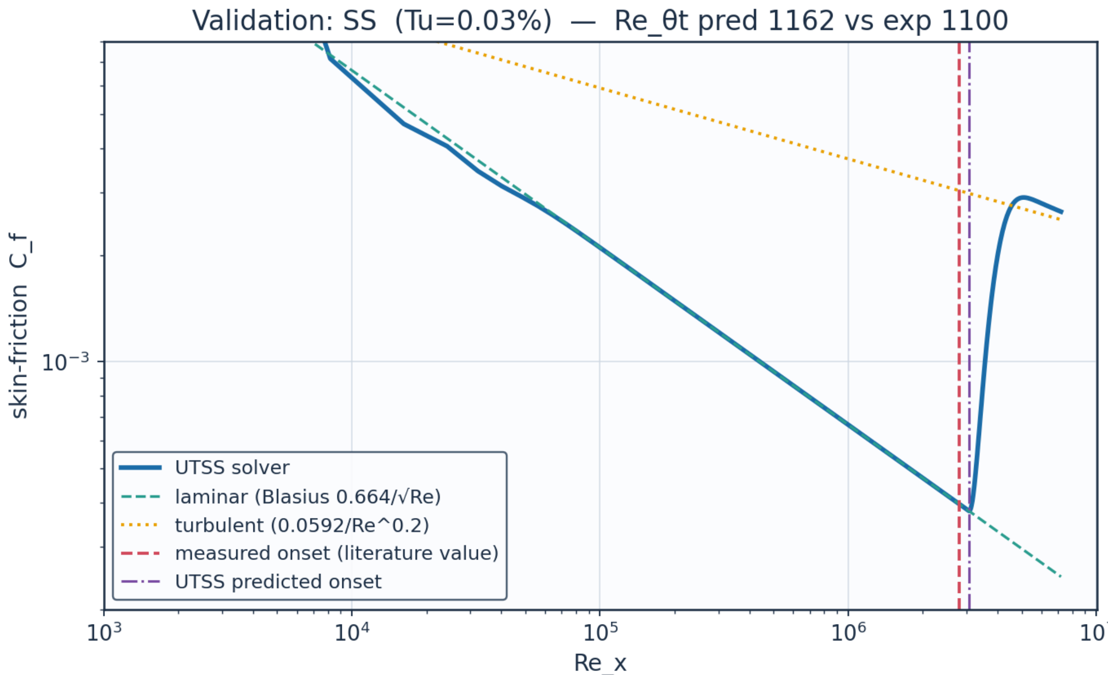
Source: Roach & Brierley (1990), ERCOFTAC T3B; Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 37. Validation — ERCOFTAC T3B flat plate (Tu = 6.0 %, bypass).



Source: Coupland J. (1990), ERCOFTAC T3C4; ERCOFTAC Classic Collection Case 020, variable-pressure-grad...

Fig. 38. Validation — ERCOFTAC T3C4 flat plate (laminar separation bubble).



Source: Schubauer & Skramstad (1948), NACA Report 909....

Fig. 39. Validation — Schubauer & Skramstad natural transition (Tu = 0.03 %).

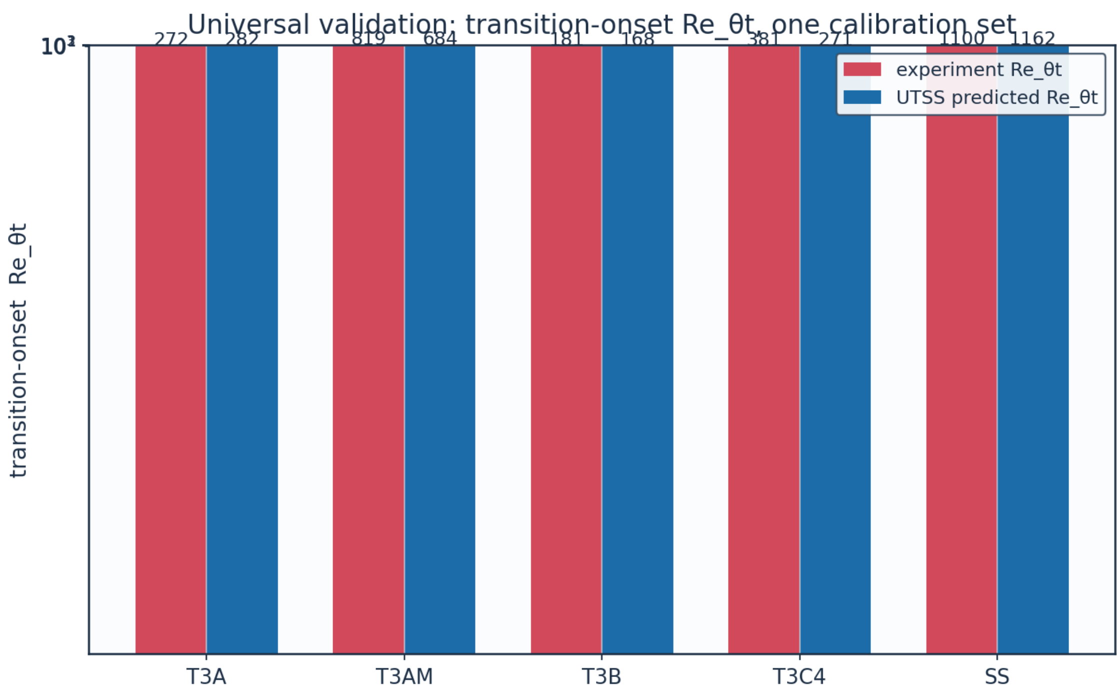


Fig. 40. Universal validation — Re_ϑt across all five plates.

11.2 NLF(1)-0416 aerofoil — 86 transition locations

The largest single body of evidence in this work, and the one on which nothing is calibrated. Transition locations were digitised from Fig. 9 of the source report at four chord Reynolds numbers on both surfaces, and each condition is matched by trimming the incidence to the measured lift coefficient. The experiment brackets transition between adjacent orifices 0.05c apart, so its own uncertainty is ±0.025c and a prediction inside that band cannot be distinguished from the measurement. Both the natural (TS) and the separation-induced branches are selected on this set.

set	points	mean_abs_err_c	bias_c	within_bracket	within_bracket_pct
Upper surface	46	0.0409	0.0025	25	54.3
Lower surface	40	0.0332	-0.0148	27	67.5
All	86	0.0373	-0.0055	52	60.5
Conditions the method accepts	83	0.0288	-0.0031	52	62.7

Table 21. NLF(1)-0416 error statistics by surface (aerofoil_nlf0416_summary.csv).

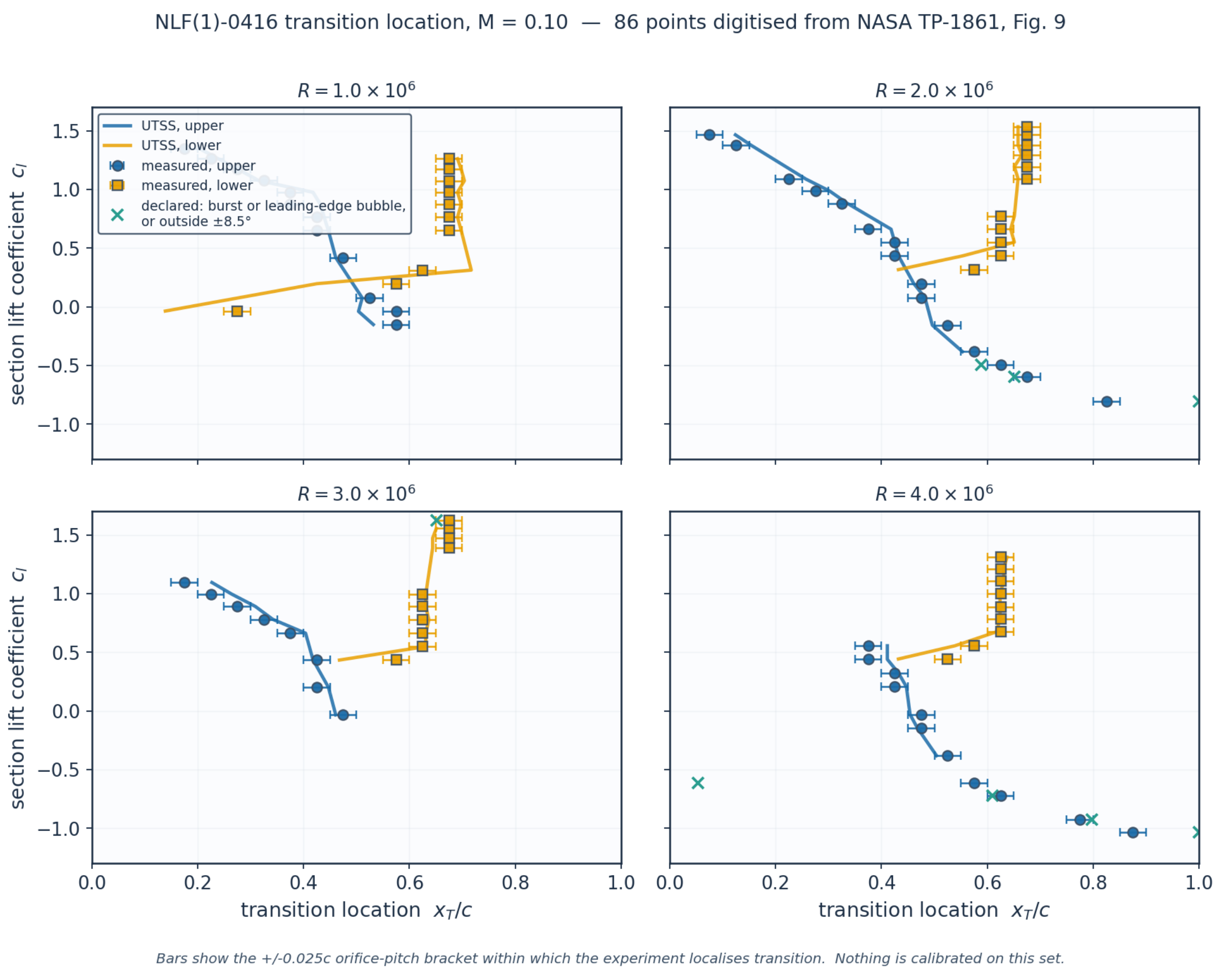


Fig. 41. Transition location against lift coefficient at four chord Reynolds numbers, measurement bars = the ±0.025c orifice bracket.

Re_c	surface	c_l_exp	alpha_deg	c_l_solver	x_tr_c_exp	x_tr_c_pred
1.0e+06	upper	1.35	6.483	1.3503	0.175	0.203
1.0e+06	upper	1.174	5.036	1.1743	0.275	0.276
1.0e+06	upper	0.976	3.413	0.9762	0.375	0.418
1.0e+06	upper	0.765	1.687	0.765	0.425	0.439
1.0e+06	upper	0.419	-1.132	0.4187	0.475	0.461
1.0e+06	upper	-0.039	-4.846	-0.0387	0.575	0.504
1.0e+06	lower	1.262	5.759	1.2623	0.675	0.691
1.0e+06	lower	1.075	4.224	1.0753	0.675	0.704
1.0e+06	lower	0.874	2.578	0.8741	0.675	0.698
1.0e+06	lower	0.651	0.757	0.6508	0.675	0.698
1.0e+06	lower	0.197	-2.935	0.1968	0.575	0.425
2.0e+06	upper	1.466	7.438	1.4661	0.075	0.124
2.0e+06	upper	1.089	4.339	1.0893	0.225	0.257
2.0e+06	upper	0.883	2.652	0.8831	0.325	0.335
2.0e+06	upper	0.551	-0.058	0.5507	0.425	0.425
2.0e+06	upper	0.198	-2.927	0.1978	0.475	0.461
2.0e+06	upper	-0.156	-5.791	-0.1552	0.525	0.496
2.0e+06	upper	-0.491	-8.514	-0.491	0.625	0.589
2.0e+06	upper	-0.807	-11.077	-0.8069	0.825	1.0
2.0e+06	lower	1.469	7.463	1.4691	0.675	0.658
2.0e+06	lower	1.294	6.022	1.2943	0.675	0.665
2.0e+06	lower	1.093	4.372	1.0933	0.675	0.658
2.0e+06	lower	0.663	0.855	0.6628	0.625	0.644
2.0e+06	lower	0.434	-1.01	0.4337	0.625	0.553
3.0e+06	upper	1.094	4.38	1.0943	0.175	0.227
3.0e+06	upper	0.89	2.709	0.8901	0.275	0.309
3.0e+06	upper	0.662	0.847	0.6618	0.375	0.404
3.0e+06	upper	0.202	-2.894	0.2018	0.425	0.447
3.0e+06	lower	1.623	8.733	1.6227	0.675	0.651
3.0e+06	lower	1.472	7.487	1.4721	0.675	0.644

(columns 2-7 of 15; Re_c repeated on each block)

Re_c	err_c	err_pct	within_bracket	degenerate	burst	le_sep
1.0e+06	0.028	16.1	False	False	False	False
1.0e+06	0.001	0.4	True	False	False	False
1.0e+06	0.043	11.5	False	False	False	False
1.0e+06	0.014	3.4	True	False	False	False
1.0e+06	-0.014	-3.0	True	False	False	False
1.0e+06	-0.071	-12.4	False	False	False	False
1.0e+06	0.016	2.4	True	False	False	False
1.0e+06	0.029	4.4	False	False	False	False
1.0e+06	0.023	3.4	True	False	False	False
1.0e+06	0.023	3.4	True	False	False	False
1.0e+06	-0.15	-26.0	False	False	False	False
2.0e+06	0.049	66.0	False	False	False	False
2.0e+06	0.032	14.3	False	False	False	False
2.0e+06	0.01	3.2	True	False	False	False
2.0e+06	0.0	0.1	True	False	False	False
2.0e+06	-0.014	-3.0	True	False	False	False
2.0e+06	-0.029	-5.4	False	False	False	False
2.0e+06	-0.036	-5.8	False	False	False	False
2.0e+06	0.175	21.2	False	True	True	False
2.0e+06	-0.017	-2.5	True	False	False	False
2.0e+06	-0.01	-1.5	True	False	False	False
2.0e+06	-0.017	-2.5	True	False	False	False
2.0e+06	0.019	3.1	True	False	False	False

2.0e+06	-0.072	-11.4	False	False	False	False
3.0e+06	0.052	29.5	False	False	False	False
3.0e+06	0.034	12.2	False	False	False	False
3.0e+06	0.029	7.8	False	False	False	False
3.0e+06	0.022	5.1	True	False	False	False
3.0e+06	-0.024	-3.5	True	False	False	False
3.0e+06	-0.031	-4.6	False	False	False	False

(columns 8-13 of 15; Re_c repeated on each block)

Re_c	x_sep_c	mechanism
1.0e+06		TS-natural
1.0e+06		TS-natural
1.0e+06	0.335	separation
1.0e+06	0.356	separation
1.0e+06	0.383	separation
1.0e+06		TS-natural
1.0e+06	0.582	separation
1.0e+06	0.582	separation
1.0e+06	0.575	separation
1.0e+06	0.568	separation
1.0e+06		TS-natural
2.0e+06		TS-natural
2.0e+06		TS-natural
2.0e+06		TS-natural
2.0e+06	0.369	separation
2.0e+06	0.404	separation
2.0e+06		TS-natural
2.0e+06		TS-natural
2.0e+06	0.947	none(laminar)
2.0e+06	0.589	separation
2.0e+06	0.589	separation
2.0e+06	0.582	separation
2.0e+06	0.568	separation
2.0e+06		TS-natural
3.0e+06		TS-natural
3.0e+06		TS-natural
3.0e+06	0.363	separation
3.0e+06	0.404	separation
3.0e+06	0.596	separation
3.0e+06	0.589	separation

Table 22. NLF(1)-0416 point-by-point comparison (aerofoil_nlf0416.csv, sampled). (columns 14-15 of 15; the table is split across 3 blocks so that no cell is compressed to illegibility, with Re_c repeated on each)

(table sampled to 30 rows; full data in 06_validation/aerofoil_nlf0416.csv)

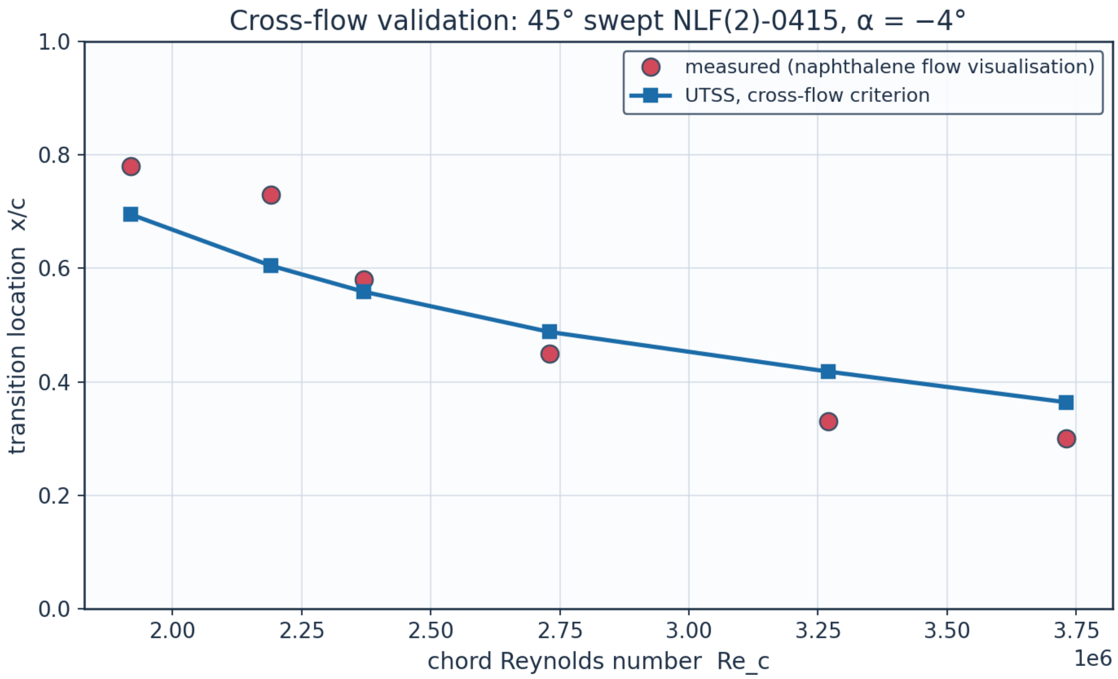
11.3 Swept wings — the cross-flow branch

The cross-flow coefficient is set on the first of these two experiments and nothing is calibrated on the second, which is a different facility, section and era. The branch reproduces the calibration set to 14.7 % at the frozen constant C1 = 150. On the independent set that same constant gives 51.7 %, and C1 = 200 gives 18.4 % — so the criterion has the right functional form on both wings but not one critical value that serves both. The independent set is therefore tabulated at BOTH ends of that band: reporting only C1 = 150 understates what the criterion does here, and reporting only C1 = 200 would be a per-case re-tune of the kind this work is claiming not to need. The spread between the two columns is the limitation, stated rather than averaged away. Nothing else in the model differs between the two columns.

What each experiment requires of the criterion is measured rather than asserted (Table 25). The march is run with every branch disabled so that it reaches the measured transition station, and the criterion is evaluated there. Dagenhart & Saric require a critical Re_θ2 of 165 with a 17.2 % coefficient of variation over six chord Reynolds numbers; Boltz et al. require 234 with 4.0 % over four sweep angles and a factor of three in chord Reynolds number. Each facility is therefore internally consistent — the second markedly so — and the two differ by 42 %. That is the shape of a receptivity difference rather than of a criterion with the wrong form: stationary cross-flow vortices are seeded by leading-edge roughness, and neither report documents the surface finish. Two attempts to close the gap fail and are recorded rather than dropped. Replacing the constant surrogate by the exact Falkner-Skan-Cooke factor K(λ) makes matters worse, taking the pooled coefficient of variation from 21 to 82 % and inverting the ratio between the two sets. Giving the cross-flow branch its own amplification threshold, separate from the one Mack's relation supplies for Tollmien-Schlichting waves — which is defensible, since a stationary cross-flow vortex is not seeded by free-stream turbulence — moves the independent set only from 55 to 51 % as that threshold is taken from N = 2 to N = 12, with C1 refitted on the calibration set at each value. The reason is that once Re_θ2 exceeds C1 the amplification builds so quickly that the threshold is nearly redundant with C1 itself, so the two constants cannot be separated by these data.

Re_c	x_tr_c_exp	x_tr_c_pred	err_pct	Re_theta_at_onset	mechanism
1.920e+06	0.78	0.695	-10.9	667.0	crossflow
2.190e+06	0.73	0.605	-17.1	662.5	crossflow
2.370e+06	0.58	0.559	-3.7	662.7	crossflow
2.730e+06	0.45	0.488	8.5	664.5	crossflow
3.270e+06	0.33	0.418	26.6	670.7	crossflow
3.730e+06	0.3	0.364	21.4	665.6	crossflow

Table 23. Cross-flow validation, 45° swept NLF(2)-0415 (Dagenhart & Saric — the calibration set).



Source: Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wi...

Fig. 42. Transition location against chord Reynolds number, 45° swept NLF(2)-0415.

sweep_deg	alpha_deg	Re_c	x_tr_c_exp	x_tr_c_pred_C1_150	err_pct_C1_150	mechanism
20.0	0.0	2.700e+07	0.207	0.093	-55.0	crossflow
30.0	0.0	1.500e+07	0.213	0.093	-56.2	crossflow
40.0	0.0	1.200e+07	0.2	0.098	-51.1	crossflow
50.0	0.0	9.500e+06	0.211	0.117	-44.4	crossflow

(columns 2-7 of 9; sweep_deg repeated on each block)

sweep_deg	x_tr_c_pred_C1_200	err_pct_C1_200
20.0	0.161	-22.4
30.0	0.161	-24.6
40.0	0.161	-19.7
50.0	0.197	-6.8

Table 24. Independent swept-wing check, NACA 64(2)A015 (Boltz et al.) at both ends of the reported cross-flow band — nothing calibrated here. (columns 8-9 of 9; the table is split across 2 blocks so that no cell is compressed to illegibility, with sweep_deg repeated on each)

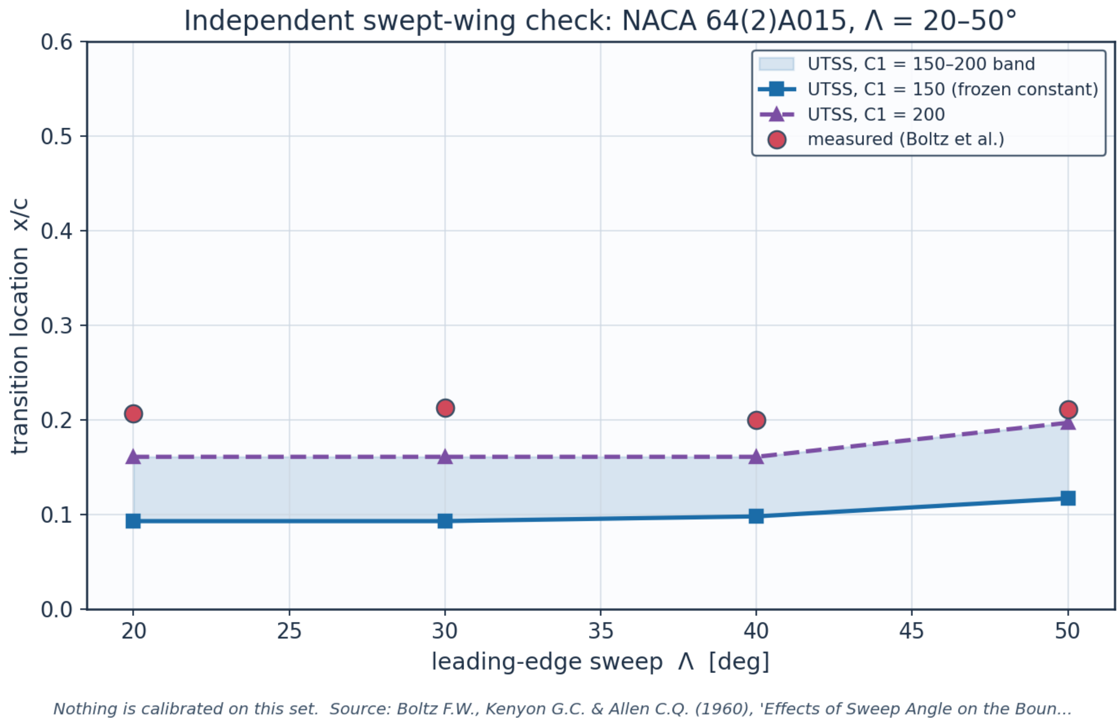


Fig. 43. Transition location against sweep angle, NACA 64(2)A015, with the C1 = 150–200 band shaded.

11.3a Where the flat-plate residuals come from

The residuals are accounted for here, and the accounting is generated by gen_validation.py rather than argued.

The laminar branch is not where they are. Compared against the model's own marched momentum thickness at the same stations — not against flat-plate Blasius, which is the wrong reference for the one plate that has a pressure gradient — the measured laminar layer agrees to within 6 % on all four plates that carry C_f data, and to 2 % on T3C4. An earlier version of this report attributed the T3C4 residual to the pre-transitional thickening a laminar layer undergoes in a turbulent free stream, on the strength of the measured momentum thickness being 1.36 times Blasius at onset. That comparison was wrong: a laminar layer in an adverse gradient is thicker than Blasius for a reason that has nothing to do with turbulence, and against the march that carries the gradient the agreement is 2 %.

The T3C4 residual is in the bubble, and Table 20b localises it. Across the dead-air region the momentum integral is exact given the edge velocity and the shape factor — there is no wall stress to get wrong — so three quantities are the whole of it. The edge velocity is verified: the smoothing spline reproduces the twelve tabulated points to 0.011 m/s, which is the precision they are quoted to, and their gradient across the bubble to 0.5 %. The shape factor is not: the march reaches 3.44 against a measured 5.17, because it is bounded by the attached Falkner-Skan branch, and that cap alone accounts for a factor 1.32 of the 2.3-fold shortfall in dθ/dx. The rest is length — the modelled bubble is 0.082 m against a measured lower bound of 0.10 m —

and the length is N_crit·θ_s/σ with all three computed rather than fitted, so there is no constant to adjust. Relaxing the shape-factor cap onto the reverse-flow profile the amplification rate is already read from, which removes an inconsistency in how the model describes the same region, moves T3C4 by 0.8 percentage points and the 86 aerofoil conditions by 0.0004c; it is exposed as cal["bub_rev_H"] and left off, because the effect is inside the discretisation scatter and the shipped setting is the better-validated one.

Conditioning. In a decaying stream the onset threshold rises while Re_θ grows only as the square root of distance, so the two curves close at a shallow angle and the crossing is sensitive. Shifting the threshold by ±10 % moves the predicted transition location by a factor of 3.5 on T3A, 2.5 on T3A⁻ and 2.0 on T3B. A correlation accurate to ten per cent cannot locate transition to ten per cent in such a flow.

case	branch	Re_theta_t_pred	Re_theta_t_exp	err_pct	laminar_run_meas_over_march	n_laminar_stations
ERCOFTAC T3A flat plate (bypass transition)	bypass	282.1	272.3	3.6	1.058	6
ERCOFTAC T3A- flat plate (low-Tu bypass)	bypass	683.6	818.8	-16.5	0.988	8
ERCOFTAC T3B flat plate (high-Tu bypass)	bypass	168.0	181.3	-7.3	1.058	4
ERCOFTAC T3C4 flat plate (laminar separation bubble)	separation	271.4	381.3	-28.8	1.02	9
Schubauer & Skramstad flat plate (natural transition)	TS-natural	1161.6	1100.0	5.6		0

(columns 2-7 of 8; case repeated on each block)

case	location_gain_bypass_threshold
ERCOFTAC T3A flat plate (bypass transition)	3.5
ERCOFTAC T3A- flat plate (low-Tu bypass)	2.5
ERCOFTAC T3B flat plate (high-Tu bypass)	2.0
ERCOFTAC T3C4 flat plate (laminar separation bubble)	
Schubauer & Skramstad flat plate (natural transition)	

Table 20a. The laminar branch against the measurements, and the sensitivity of the crossing (residual_diagnostics.csv). (columns 8-8 of 8; the table is split across 2 blocks so that no cell is compressed to illegibility, with case repeated on each)

quantity	model	measured	note
separation station x [m]	1.236	1.295	model / first station at the C_f floor

reattachment station x [m]	1.318	1.395	model / last station at the C_f floor
bubble length [m]	0.082	0.1	measured length is a lower bound
dtheta/dx across the bubble [1/m]	1.880e-03	5.906e-03	momentum integral is exact given U_e and H
shape factor at reattachment	3.44	5.17	model bounded by the attached Falkner-Skan branch, H <= 3.997
dU_e/dx over the bubble [1/s]	-0.3782	-0.38	smoothing spline against the tabulated points
max |U_e spline - tabulated| [m/s]	0.011	0.01	tabulated to 0.01 m/s, so the spline is within quotation

Table 20b. The T3C4 bubble, modelled against measured (bubble_diagnostics.csv). The momentum integral across the dead-air region is exact given U_e and H, so these are the whole of the residual.

11.4 What the two swept-wing experiments require of the criterion

dataset	criterion	mean_critical_value	coeff_of_variation_pct	n_points
Dagenhart & Saric (calibration)	surrogate Re_theta2	164.8	17.2	6
Dagenhart & Saric (calibration)	exact Falkner-Skan-Cooke Re_cf	440.4	58.9	6
Boltz et al. (independent)	surrogate Re_theta2	233.8	4.0	4
Boltz et al. (independent)	exact Falkner-Skan-Cooke Re_cf	117.8	4.8	4
Both facilities pooled	surrogate Re_theta2	192.4	21.2	10
Both facilities pooled	exact Falkner-Skan-Cooke Re_cf	311.4	82.1	10

Table 25. What each swept-wing experiment requires of the cross-flow criterion, evaluated at the measured transition station (crossflow_criticals_summary.csv).

dataset	sweep_deg	Re_c	x_tr_c_measured	Re_theta	Re_theta2_surrogate	Re_cf_exact_FSC
Dagenhart & Saric (calibration)	45.0	1.920e+06	0.78	935.7	219.9	872.9
Dagenhart & Saric (calibration)	45.0	2.190e+06	0.73	769.7	180.9	718.0
Dagenhart & Saric (calibration)	45.0	2.370e+06	0.58	675.0	158.6	351.8
Dagenhart & Saric (calibration)	45.0	2.730e+06	0.45	637.1	149.7	275.8
Dagenhart & Saric (calibration)	45.0	3.270e+06	0.33	590.7	138.8	215.0
Dagenhart & Saric (calibration)	45.0	3.730e+06	0.3	598.6	140.7	209.1
Boltz et al. (independent)	20.0	2.700e+07	0.207	1570.9	237.3	119.8
Boltz et al. (independent)	30.0	1.500e+07	0.213	1189.0	242.0	119.6

Boltz et al. (independent)	40.0	1.200e+07	0.2	1028.0	237.9	123.4
Boltz et al. (independent)	50.0	9.500e+06	0.211	941.4	217.9	108.4

Table 26. Point-by-point cross-flow criterion values at the measured transition stations (crossflow_criticals.csv).

One explanation can be ruled out rather than merely doubted. If the difference between the two facilities were a Reynolds-number effect the criterion is missing, the requirement would have to vary with chord Reynolds number in the same direction within a facility as it does between them. It does not. Within Dagenhart & Saric the required critical value FALLS steeply with chord Reynolds number, by 251 per decade with a correlation of −0.88 over a factor of two in Re_c; Boltz et al. sit at six times that Reynolds number and require 42 % MORE, not less, and are themselves flat across a factor of three. The between-facility offset therefore has the opposite sign to the within-facility trend, and no monotone function of Re_c can carry one set into the other. That is what leaves receptivity — the leading-edge surface finish neither report documents — as the explanation, and it is now a measurement rather than an appeal to the literature.

Two numbers in this section have to be reconciled or they look inconsistent: the band is reported as C1 = 150–200, while Table 25 says the independent facility requires a critical Re_θ2 of 234. They are different quantities. C1 does not fire the branch; it starts the amplification integral, which then decides where transition is placed, so the EFFECTIVE critical value at the predicted station is higher than C1. Measured at the four Boltz conditions it is 157 for C1 = 150 and 209 for C1 = 200 — the integral adds about 5 %, not the 17 % that would be needed to reach 234. That residual 12 % is precisely why the upper end of the band still leaves an 18.4 % error in location on the independent set, and it is the honest reason the band is offered as a bound rather than as a value.

dataset	n_points	Re_c_min	Re_c_max	mean_ required	slope_per_ decade_Re_c	correlation
Dagenhart & Saric (calibration)	6	1.92e+06	3.73e+06	164.8	-251.2	-0.881
Boltz et al. (independent)	4	9.50e+06	2.70e+07	233.8	32.1	0.58

Table 26a. The required critical value against chord Reynolds number within each facility (crossflow_reynolds_trend.csv). The trends have opposite signs to the offset between the facilities.

11.5 Ablations — what each element of the formulation is worth

Each of the three elements that distinguish this formulation is switched off in turn, everything else held fixed, and the same two datasets the natural branch reaches are re-run. The table is regenerated by gen_validation.py and is not quoted from memory.

configuration	SS_err_pct	mean_abs_err_c	within_bracket	points	within_bracket_ pct
no bubble closure	5.6	0.0607	19	86	22.1
one-equation	-4.7	0.0388	44	86	51.2

laminar march					
Drela-Giles envelope	-6.6	0.0451	39	86	45.3
full model	5.6	0.0373	52	86	60.5

Table 27. Ablation study (ablations.csv): Schubauer-Skramstad onset error and the 86 aerofoil conditions, one closure removed at a time.

Calibration. The solver is calibrated through the single constant set of Table 7. The critical amplification factor is not among the constants: it follows from the free-stream turbulence intensity by Mack's correlation, clamped to the 0.0008-0.0298 range over which that correlation is quoted, and so returns 8.18 for any stream quieter than 0.08 %. The SAME constants reproduce every validation dataset — five flat plates, two swept wings and 86 aerofoil conditions — demonstrating that no case-specific physics tuning is required, which is the essence of the universality claim.

11.6 Sources and references

All data sources used for validation, for calibration of the closures, and for the case-study definition are recorded below.

category	item	reference
Validation	ERCOFTAC T3A flat plate	Roach P.E. & Brierley D.H. (1990), 'The influence of a turbulent free stream on zero pressure gradient transitional boundary layer development', ERCOFTAC Workshop; reproduced in Savill (1993) and Langtry & Menter, AIAA J. 47(12), 2009.
Validation	ERCOFTAC T3B flat plate	Roach & Brierley (1990), ERCOFTAC T3B (Tu=6%); Langtry & Menter, AIAA J. 47(12), 2009.
Validation	Schubauer & Skramstad natural transition	Schubauer G.B. & Skramstad H.K. (1948), 'Laminar boundary-layer oscillations and transition on a flat plate', NACA Report 909.
Calibration	Bypass onset correlation (AGS)	Abu-Ghannam B.J. & Shaw R. (1980), 'Natural transition of boundary layers - the effects of turbulence, pressure gradient and flow history', J. Mech. Eng. Sci. 22(5).
Calibration	Laminar integral method	Thwaites B. (1949), 'Approximate calculation of the laminar boundary layer', Aero. Quarterly 1.
Calibration	Turbulent entrainment / Cf	Head M.R. (1958) entrainment method; Ludwieg H. & Tillmann W. (1950) skin-friction law.
Calibration	Intermittency / transition length	Narasimha R. (1957); Dhawan S. & Narasimha R. (1958), J. Fluid Mech. 3.
Calibration	Cross-flow criterion	Arnal D. (1984), C1 cross-flow criterion; Mayle R.E. (1991), J. Turbomachinery 113.
Validation	ERCOFTAC T3A- flat plate	Roach & Brierley (1990), ERCOFTAC T3A- (Tu=0.87%); ERCOFTAC Classic Collection Case 020.
Validation	ERCOFTAC T3C4 flat plate (separation bubble)	Coupland J. (1990), ERCOFTAC T3C4; ERCOFTAC Classic Collection Case 020,

		variable-pressure-gradient series.
Validation	NLF(1)-0416 aerofoil transition (86 conditions)	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861, Fig. 9; tunnel turbulence level from McGhee R.J., Beasley W.D. & Foster J.M. (1984), NASA TP-2328, Fig. 25.
Validation	Swept-wing cross-flow (calibration set)	Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wing Flow', NASA/TP-1999-209344, Table 2.
Validation	Swept-wing cross-flow (independent set)	Boltz F.W., Kenyon G.C. & Allen C.Q. (1960), 'Effects of Sweep Angle on the Boundary-Layer Stability Characteristics of an Untapered Wing at Low Speeds', NACA TN D-338, Fig. 9(g).
Calibration	Linear stability / amplification database	Orr (1907); Sommerfeld (1908); Gaster M. (1962), JFM 14, 222; Mack L.M. (1977), AGARD CP-224; Drela M. & Giles M.B. (1987), AIAA J. 25(10), 1347.
Case study	NLF aerofoil design reference	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861 (NLF(1)-0416).
Case study	Panel method	Kuethe A.M. & Chow C.Y. (1998), 'Foundations of Aerodynamics', 5th ed., Wiley.
Case study	ISA atmosphere	ICAO Standard Atmosphere (ISO 2533:1975), FL360 properties.

Table 28. Validation, calibration and case-study sources.

dataset	source
Dagenhart & Saric 45 deg swept NLF(2)-0415 (cross-flow)	Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wing Flow', NASA/TP-1999-209344, Table 2.

Table 29. Cross-flow calibration dataset provenance.

dataset	source
Boltz, Kenyon & Allen NACA 64(2)A015 untapered wing	Boltz F.W., Kenyon G.C. & Allen C.Q. (1960), 'Effects of Sweep Angle on the Boundary-Layer Stability Characteristics of an Untapered Wing at Low Speeds', NACA TN D-338, Fig. 9(g); transition locations digitised by the present author as x/c = R_trans/R.

Table 30. Independent swept-wing dataset provenance.

dataset	source	n_points	Tu_pct	orifice_pitch
Somers NLF(1)-0416 aerofoil, Langley LTPT	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861, Fig. 9 (transition) and Table	86	0.03	0.05

	I (coordinates); turbulence level from McGhee R.J., Beasley W.D. & Foster J.M. (1984), 'Recent Modifications and Calibration of the Langley Low-Turbulence Pressure Tunnel', NASA TP-2328, Fig. 25.			

Table 31. Aerofoil dataset provenance.

12. Contribution to Knowledge

- A single unified transition kernel (Eq. E13) that locally selects the governing mechanism among natural-TS, bypass, separation-induced and cross-flow transition by a minimum-effective-onset rule — reproducing all regimes with one calibration set.
- An amplification database in place of an envelope correlation: 61,600 Orr-Sommerfeld eigenvalue solutions on the Falkner-Skan family, tabulated against shape factor, momentum-thickness Reynolds number and frequency, with the march carrying one amplification factor per physical frequency. The stability solver reproduces the Blasius neutral point at Re_theta = 201 against the accepted 200.5.
- A separation-bubble closure that predicts a length rather than a point: the shear layer is carried across the dead-air region by the momentum integral with no wall stress — a step with no fitted constant, which reproduces the growth the T3C4 hot films record — and reattachment placed where the disturbance has amplified by the same N_crit used elsewhere, so the length scales with the disturbance environment.
- A two-equation laminar march whose closures are computed from the Falkner-Skan family rather than fitted, giving the shape factor a history. On the 86 aerofoil conditions it raises the number of predictions inside the experimental bracket from 44 to 52, improving both surfaces at once.
- Regime coverage with one constant set: transition-onset Re_theta_t predicted to +3.6 % on T3A, -16.5 % on T3A-, -7.3 % on T3B, -28.8 % on T3C4, +5.6 % on Schubauer & Skramstad, spanning 0.03-6 % free-stream turbulence intensity.
- An aerofoil validation built for this work: 86 transition locations digitised from Fig. 9 of NASA TP-1861 for the NLF(1)-0416 section, both surfaces, four chord Reynolds numbers and lift coefficients from -1.03 to +1.62, with nothing calibrated on them. Mean error 0.037 chord, and 52 of the 86 predictions fall inside the +/-0.025c bracket within which the experiment itself localises transition.
- A robust, panel-method-cost (<1 s) 3-D capability via a span-wise strip formulation with built-in cross-flow, suitable for design-loop use where RANS/LES are impractical.
- An end-to-end, auditable workflow (geometry → mesh → setup → solution → post-processing → validation) producing a complete engineering output set (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors).

- Quantified NLF benefit for the case vehicle: 57 % laminar flow and 50 % viscous drag reduction relative to a fully-turbulent wing at the cruise design point.

13. Conclusions

The UTSS universal transition & skin-friction solver predicts boundary-layer transition over the three-dimensional NLF wing of the AETHER-NLF 25 and the dependent engineering quantities (skin friction, laminar extent, boundary-layer growth, separation margin, profile drag) at panel-method cost. The unified four-mechanism kernel, validated with a single calibration set against five flat plates, two independent swept-wing experiments and 86 aerofoil conditions, spans 0.03-6 % free-stream turbulence intensity. At cruise the wing achieves 57 % laminar flow and a 50 % viscous-drag reduction versus a turbulent wing; at the higher-turbulence climb condition the solver switches to the bypass route and predicts early transition, demonstrating regime coverage across the flight envelope.

Two limitations bound that claim and are stated here rather than left to be discovered. The cross-flow critical constant does not transfer between facilities: the two independent swept-wing experiments support the functional form of the criterion, and the measured data collapse on it, but reproducing the second requires a critical value 42 % larger than the first. And the method is an attached-flow formulation with an incidence envelope of about +/-8.5 deg on a real section; beyond that the laminar march separates near the leading edge and the result should not be relied on.

Appendix A. Complete Generated-Output Inventory

Every file generated for this case study, by folder:

Total generated data/figure files: 111 (CSV: 50, PNG: 59, NPZ: 2).

file	type
01_geometry/airfoil_UTSS-NLF16.csv	CSV
01_geometry/geometry_definition.csv	CSV
01_geometry/wing_planform.csv	CSV
01_geometry/wing_sections_3d.csv	CSV
01_geometry/drawings/dwg_01_airfoil_section.png	PNG
01_geometry/drawings/dwg_02_planview.png	PNG
01_geometry/drawings/dwg_03_front_side.png	PNG
01_geometry/drawings/dwg_04_orthographic.png	PNG
01_geometry/drawings/dwg_05_isometric.png	PNG
01_geometry/drawings/dwg_06_section_BB.png	PNG
02_mesh/bl_normal_grid.csv	CSV
02_mesh/mesh_independence.csv	CSV
02_mesh/mesh_metrics.csv	CSV
02_mesh/surface_mesh_nodes.csv	CSV
02_mesh/plots/mesh_01_surface.png	PNG
02_mesh/plots/mesh_02_bl_normal.png	PNG
02_mesh/plots/mesh_03_independence.png	PNG

03_model_setup/calibration_constants.csv	CSV
03_model_setup/flow_conditions.csv	CSV
03_model_setup/material_properties.csv	CSV
03_model_setup/solver_settings.csv	CSV
04_solution/aero_polar.csv	CSV
04_solution/bl_profiles_cruise.csv	CSV
04_solution/field_pressure_climb.csv	CSV
04_solution/field_pressure_climb.npz	NPZ
04_solution/field_pressure_cruise.csv	CSV
04_solution/field_pressure_cruise.npz	NPZ
04_solution/integrated_forces.csv	CSV
04_solution/nlf_vs_turbulent.csv	CSV
04_solution/spanwise_distribution.csv	CSV
04_solution/surface_climb_lower.csv	CSV
04_solution/surface_climb_upper.csv	CSV
04_solution/surface_cruise_lower.csv	CSV
04_solution/surface_cruise_upper.csv	CSV
04_solution/transition_length_sensitivity.csv	CSV
04_solution/transition_summary.csv	CSV
05_postprocessing/csv_plots/aero_polar.png	PNG
05_postprocessing/csv_plots/calibration_constants.png	PNG
05_postprocessing/csv_plots/climb_Cf.png	PNG
05_postprocessing/csv_plots/climb_Cp.png	PNG
05_postprocessing/csv_plots/climb_Retheta.png	PNG
05_postprocessing/csv_plots/climb_gamma.png	PNG
05_postprocessing/csv_plots/climb_theta_H.png	PNG
05_postprocessing/csv_plots/compare_cruise_climb_Cf.png	PNG
05_postprocessing/csv_plots/conditions_compare.png	PNG
05_postprocessing/csv_plots/cruise_Cf.png	PNG
05_postprocessing/csv_plots/cruise_Cp.png	PNG
05_postprocessing/csv_plots/cruise_Retheta.png	PNG
05_postprocessing/csv_plots/cruise_gamma.png	PNG
05_postprocessing/csv_plots/cruise_theta_H.png	PNG
05_postprocessing/csv_plots/geo_airfoil.png	PNG
05_postprocessing/csv_plots/geo_planform.png	PNG
05_postprocessing/csv_plots/geo_sections_3d.png	PNG
05_postprocessing/csv_plots/mesh_independence.png	PNG
05_postprocessing/csv_plots/mesh_spacing.png	PNG
05_postprocessing/csv_plots/mesh_yplus.png	PNG
05_postprocessing/csv_plots/nlf_vs_turbulent.png	PNG
05_postprocessing/csv_plots/spanwise_transition.png	PNG
05_postprocessing/csv_plots/table_geometry_definition.png	PNG
05_postprocessing/csv_plots/table_integrated_forces.png	PNG
05_postprocessing/csv_plots/table_material_properties.png	PNG
05_postprocessing/csv_plots/table_mesh_metrics.png	PNG
05_postprocessing/csv_plots/table_solver_settings.png	PNG
05_postprocessing/csv_plots/transition_summary_bar.png	PNG
05_postprocessing/three_d/td_Cf.png	PNG
05_postprocessing/three_d/td_Cp.png	PNG
05_postprocessing/three_d/td_gamma.png	PNG
05_postprocessing/three_d/td_skinfriction_vectors.png	PNG
05_postprocessing/contours/contour_Cp_climb.png	PNG
05_postprocessing/contours/contour_Cp_cruise.png	PNG
05_postprocessing/contours/contour_speed_climb.png	PNG
05_postprocessing/contours/contour_speed_cruise.png	PNG
05_postprocessing/contours/vectors_climb.png	PNG

05_postprocessing/contours/vectors_cruise.png	PNG
05_postprocessing/profiles/bl_temperature_profiles.png	PNG
05_postprocessing/profiles/bl_temperature_ratio.png	PNG
05_postprocessing/profiles/bl_velocity_profiles.png	PNG
06_validation/ablations.csv	CSV
06_validation/aerofoil_nlf0416.csv	CSV
06_validation/aerofoil_nlf0416_source.csv	CSV
06_validation/aerofoil_nlf0416_summary.csv	CSV
06_validation/bubble_diagnostics.csv	CSV
06_validation/crossflow_criticals.csv	CSV
06_validation/crossflow_criticals_summary.csv	CSV
06_validation/crossflow_reynolds_trend.csv	CSV
06_validation/experiment_T3A.csv	CSV
06_validation/experiment_T3AM.csv	CSV
06_validation/experiment_T3B.csv	CSV
06_validation/experiment_T3C4.csv	CSV
06_validation/residual_diagnostics.csv	CSV
06_validation/solver_SS.csv	CSV
06_validation/solver_T3A.csv	CSV
06_validation/solver_T3AM.csv	CSV
06_validation/solver_T3B.csv	CSV
06_validation/solver_T3C4.csv	CSV
06_validation/sources_and_references.csv	CSV
06_validation/swept_wing_crossflow.csv	CSV
06_validation/swept_wing_independent.csv	CSV
06_validation/swept_wing_independent_source.csv	CSV
06_validation/swept_wing_source.csv	CSV
06_validation/validation_summary.csv	CSV
06_validation/plots/val_SS.png	PNG
06_validation/plots/val_T3A.png	PNG
06_validation/plots/val_T3AM.png	PNG
06_validation/plots/val_T3B.png	PNG
06_validation/plots/val_T3C4.png	PNG
06_validation/plots/val_aerofoil_nlf0416.png	PNG
06_validation/plots/val_combined_Re_theta_t.png	PNG
06_validation/plots/val_swept_crossflow.png	PNG
06_validation/plots/val_swept_independent.png	PNG
07_equations/equations_index.csv	CSV

Table A1. Generated-output inventory.

Appendix B. Parameter / Metric CSVs Rendered as Figures

For completeness, every remaining parameter and metric CSV is also rendered as a clean figure so that all generated CSV data appear in plotted form (the same data also appear as native tables earlier).

Geometry definition (geometry_definition.csv)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m^2
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section c_l at cruise design incidence (alpha = 1.5 deg, M = 0.42)	0.517	-

Fig. B1. geometry_definition.csv.

Mesh metrics (mesh_metrics.csv)

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y1	6.51	micron
Target wall y+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	1e-3 c
Max streamwise spacing	9.927	1e-3 c
Max streamwise spacing at MAC	20.074	mm
LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_tau (TE)	4.800	m/s
Max cell aspect ratio (at MAC)	3082	-
Discretisation type	surface panel distribution + wall-normal reconstruction stack (no volume mesh is generated)	-

Fig. B2. mesh_metrics.csv.

Air / material properties (material_properties.csv)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m^3

Fig. B3. material_properties.csv.

Solver configuration (solver_settings.csv)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility (pressure)	Karman-Tsien correction on C_p
Compressibility (boundary layer)	closures evaluated at Eckert's reference temperature
Laminar BL closure	two-equation march (momentum + kinetic energy); H*, I and d from the Falkner-Skan family
Transition model	UTSS unified 4-mechanism kernel
TS / natural	e^N, one factor per physical frequency, growth rates from the tabulated Orr-Sommerfeld database
bypass	Abu-Ghannam & Shaw at the flow-history-averaged Tu
separation	laminar bubble closed by the amplification integral across the dead-air region
cross-flow	C1 on Re_theta2, closed by an amplification integral
Intermittency closure	Narasimha universal (gamma)
Transition length	Dhawan & Narasimha, Re_lambda = 9 Re_x_t^0.75
Turbulent BL closure	Head entrainment + Ludwieg-Tillmann Cf
Laminar separation criterion	Thwaites lambda <= -0.09
Turbulent separation flag	H > 2.6
Drag integration	Squire-Young far-wake, evaluated at 0.98c
Marching scheme	explicit trapezoidal, arc-length stepping with sub-stepping on the kinetic-energy equation
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	~1 (reconstruction grid)

Fig. B4. solver_settings.csv.

Integrated forces (integrated_forces.csv)

quantity	value	unit
Section lift coefficient Cl	0.5171	-
Section profile drag Cd	0.00437	-
Section Cd (counts)	43.7	counts
Section L/D	118.2	-
Section lift-curve slope a0 (panel)	8.111	1/rad
Zero-lift incidence a_L0 (panel)	-2.185	deg
Quarter-chord sweep	9.89	deg
Wing C_L (lifting line, taper + washout + sweep)	0.2548	-
Wing C_L / section c_l	0.493	-
Span efficiency e (lifting line)	0.8718	-
Wing induced drag C_Di (lifting line)	0.00272	-
Wing lift (lifting line)	23603.0	N
Wing C_L for level flight at MTOW	0.8998	-
Incidence for that C_L (lifting line)	7.55	deg
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6.414e+06	-
Mach number	0.42	-

Fig. B5. integrated_forces.csv.